Machine Learning for Energy Calibration in High Energy Physics Experiments Using the *Lorenzetti Showers Framework*

Paulo R. A. da Silva, Eduardo F. Simas Filho, Paulo C. M. A. Farias, Edmar E. P. de Souza, Juan L. Marin, and José M. Seixas

***Abstract*—Accurate energy calibration is essential in high-energy physics (HEP) experiments to improve trigger-level decision-making and enhance the selection of events involving electron and photon final states. Raw calorimetric measurements are often affected by pile-up, electronic noise, and detector non-linearities, introducing systematic biases that impair signal quality. This work addresses these challenges using simulated data generated by the Lorenzetti Showers framework. We evaluate machine learning (ML) models for estimating event-by-event calibration factors, employing compact and efficient feature sets obtained via correlation, mutual information, principal component analysis (PCA), and independent component analysis (ICA) techniques. Gradient-boosted decision trees (BDTs) without binning achieved the best performance, with up to 58.4% reduction in root mean squared error (RMSE) and 35.2% reduction in interquartile range (IQR) compared to the standard method. These gains were obtained using fewer than ten input features and with prediction times under 10 ms on standard hardware, indicating feasibility for deployment in low-latency environments. Statistical tests confirmed that the improvements in both mean accuracy and variance of calibration errors are statistically significant. The proposed ML-based strategies offer a promising solution for real-time energy calibration in trigger systems, improving the precision of online event reconstruction in future collider experiments.**

***Index Terms*—Energy calibration, feature extraction, high energy physics (HEP), Lorenzetti Showers, machine learning (ML).**

Received 22 August 2025; revised 21 November 2025; accepted 17 April 2026. Date of publication 21 April 2026; date of current version 19 May 2026. This work was supported in part by Bahia State Research Support Foundation Fundação de Amparo à Pesquisa do Estado da Bahia (FAPESB), in part by Bahia State Research Support Foundation Fundação de Amparo à Pesquisa do Estado da Bahia (FAPESB), in part by National Council for Scientific and Technological Development Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), in part by Brazilian National Network for High Energy Physics Rede Nacional de Física de Altas Energias (RENAFAE), in part by the French Committee for the Evaluation of University and Scientific Cooperation with Brazil Comité Français d'Évaluation de la Coopération Universitaire et Scientifique avec le Brésil (COFECUB), and in part by the Coordination for the Improvement of Higher Education Personnel Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), under Grant 001. *(Corresponding author: Paulo R. A. da Silva.)*

Paulo R. A. da Silva, Eduardo F. Simas Filho, Paulo C. M. A. Farias, and Edmar E. P. de Souza are with the Electrical and Computer Engineering Department, Federal University of Bahia, Salvador 40210-630, Brazil (e-mail: araujop@ufba.br; eduardo.simas@ufba.br; paulo.farias@ufba.br; edmar.egidio@ufba.br).

Juan L. Marin is with the Federal Institute of Education, Science and Technology of Bahia, Salvador 40301-015, Brazil (e-mail: juan.marin@ifba.edu.br).

José M. Seixas is with the LPS/COPPE/Polytechnic School, Federal University of Rio de Janeiro, Rio de Janeiro 21941-972, Brazil (e-mail: seixas@lps.ufrj.br).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TNS.2026.3686269.

Digital Object Identifier 10.1109/TNS.2026.3686269

I. Introduction

DETECTING and characterizing rare matter constituents pose significant challenges in particle physics, often requiring high-energy collisions at large-scale accelerators [1]. High-energy physics (HEP) experiments are essential for identifying and understanding the fundamental components of matter. In particular, accurate calibration of online and trigger-level systems for electron and photon signals plays a key role in analyses involving bosons such as the Higgs boson [2]. These experiments, however, generate vast volumes of data, much of which comprises background noise (nonrelevant events). It is important to note that only a fraction of the recorded events contain final states of interest, and their signals are often affected by the collision environment and detector inefficiencies. Therefore, a well-calibrated and efficient trigger system is crucial for suppressing irrelevant events and enhancing the selection of those most relevant to the intended analyses.

At the LHC, for instance, collision energies and luminosities have increased substantially between 2011 and 2024 [3], [4], leading to a corresponding rise in data volume and complexity. This includes pile-up effects from multiple simultaneous collisions, electronic noise, and intrinsic nonlinearities in the detector response [5], all of which hinder the accurate identification of rare physics processes.

Efficient event selection and precise energy reconstruction are thus indispensable for managing the overwhelming data influx in modern accelerators [6], [7]. This study focuses on improving energy estimation for electrons and photons, particles commonly produced in decays involving Higgs, Z, and W bosons [8]. High-precision energy estimates are not only essential for correct classification of physical processes but also contribute to minimizing storage and transmission demands [9], [10]. In this context, machine learning (ML) techniques have emerged as powerful tools to enhance energy calibration in calorimetric trigger systems.

Specifically, recent advances in ML have demonstrated the potential of tree-based ensemble methods such as gradient boosted decision trees (BDTs) to improve energy estimation in HEP detectors [11], [12], [13]. These approaches leverage high-dimensional calorimeter data and learn non-linear mappings that outperform conventional rule-based calibration. In this study, BDTs are applied in conjunction with the Ringer algorithm [8], [14], which condenses signals from over 1000 calorimeter cells into 100 concentric energy rings, facilitating fast and memory-efficient feature encoding.

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

In this work, we extend these contributions by applying feature selection and extraction techniques to the 100 energy rings in order to further reduce dimensionality and improve calibration performance. Our goal is to identify compact and informative subsets of features that simultaneously enhance energy accuracy, decrease storage costs, and reduce processing latency.

Unlike previous studies, we employ the Lorenzetti Showers framework [15], [16], a general-purpose calorimeter simulation tool, to generate the dataset used in this work. For the present study, the framework was configured to reproduce the geometry, materials, and segmentation of the ATLAS experiment electromagnetic calorimeter [17]. This configuration was chosen as a representative use case, given the extensive public documentation of the ATLAS detector and its relevance in HEP, but it does not correspond to the official ATLAS simulation framework. The adoption of Lorenzetti provides important advantages: 1) straightforward execution of simulations with flexible configurations; 2) full access to low-level calorimeter information, such as cell readings, instrumentation signals, and calorimeter rings, enabling the development and testing of ML algorithms; and 3) the possibility of generating open datasets with detailed instrumentation-level information, which are usually restricted to members of experimental collaborations. While ATLAS serves as the reference example in this article, the same methodology can be extended to other calorimetric systems by reconfiguring the Lorenzetti framework.

The main contribution of this study is the development of a robust methodology for energy calibration in HEP experiments, addressing the demand for precision under computational constraints. The proposed approach incorporates detailed detector simulation with tailored binning strategies and systematically evaluates feature selection and dimensionality reduction techniques applied to ring-encoded inputs. We compare several ML regression models, both with and without binning, to identify the most effective solutions in terms of calibration accuracy, inference time, and resource efficiency. These results demonstrate the pivotal role of optimized calibration pipelines in enhancing physics analyses in modern HEP environments.

This article is organized as follows. Section II describes the simulation setup and the complete methodology for fast energy calibration, including the Lorenzetti Showers framework, ring-based feature generation, and the calibration strategy, as well as the feature selection and extraction techniques employed to reduce dimensionality while preserving physical relevance, and details the ML models and regression strategies used for estimating calibration factors. Section III provides a comprehensive evaluation of the methods, analyzing performance metrics such as root mean squared error (RMSE) and interquartile range (IQR), as well as computational efficiency in terms of feature count and inference time. Section IV concludes this article with a summary of key findings and outlines directions for future developments.

II. Methodology

This study employs the Lorenzetti Showers framework to simulate a general-purpose segmented calorimeter system [15], [16]. All simulations were performed exclusively within Lorenzetti, using version 2.0.0, ensuring full reproducibility of the experimental setup. The framework is a self-contained tool that encapsulates event generation, shower propagation in a pre-defined detector geometry, and reconstruction of relevant calorimetric variables. It also allows the simulated electronic response of the calorimeter system to be easily modified. The installation setup of Lorenzetti follows a container-like structure that automatically includes the latest available versions of external packages at the time of release. In the case of Lorenzetti 2.0.0, this corresponds to Pythia 8.3 and Geant4 11.2.2. References to Geant4 [18] and Pythia8 [19] are therefore included solely to acknowledge the underlying technologies on which Lorenzetti was originally built and validated, as documented in [15], and not to indicate that these environments were run independently in the present work.

The Lorenzetti Showers framework, employed in this study, has been comprehensively validated by the developer community and utilized in various studies, including online energy estimation in dedicated field-programmable gate array (FPGA) hardware [20] and anomaly detection in time series [21]. A previous publication by our research group [11] has already addressed the simulation configuration used in this work. In that study, single electron events (e^+ and e^-) generated with a particle gun were produced over a wide energy range, from a few GeV up to approximately 500 GeV, and the Lorenzetti simulated calorimeter response was quantitatively compared against experimental data, showing agreement across key observables, including energy deposition profiles, longitudinal and transverse shower shapes, layer-by-layer response, energy resolution profile, and pileup effects. The energy resolution curve of the simulated calorimeter was also estimated, showing consistency with experimental data. This validation demonstrates that the framework provides a physically accurate and experimentally consistent description of electromagnetic showers, thereby ensuring the reliability of the simulated datasets used in this study. The exact electromagnetic physics configuration, energy thresholds, and geometry parameters adopted in this study are identical to those validated in [11], thus preserving the same level of physical accuracy and experimental consistency.

The use of the Lorenzetti Shower framework enables access to low-level calorimeter information in simulated data, as the electronics pulse-shaping and data acquisition circuits are emulated using analytical descriptions of these processes. In this way, the calibration studies conducted here utilize data very close to experimental ones. Additionally, to openly share the generated dataset broadens the scientific scope of this work, as usually, such data are generally unavailable outside experimental collaborations.

In this work, approximately 300 000 single-electron events from proton-proton collisions were generated within Lorenzetti using its built-in event generation pipeline. The simulation assumed a luminosity of $10^{34}\,\mathrm{cm}^{-2}\cdot\mathrm{s}^{-1}$, consistent with LHC Run 3 conditions [17]. The complete simulated dataset used in this study is publicly available at [22], ensuring full transparency and reproducibility of the results.

Electronic noise is added during the pulse-shaping stage of the Lorenzetti simulation, which converts simulated hits into readout signals, emulating real calorimeter electronics. This

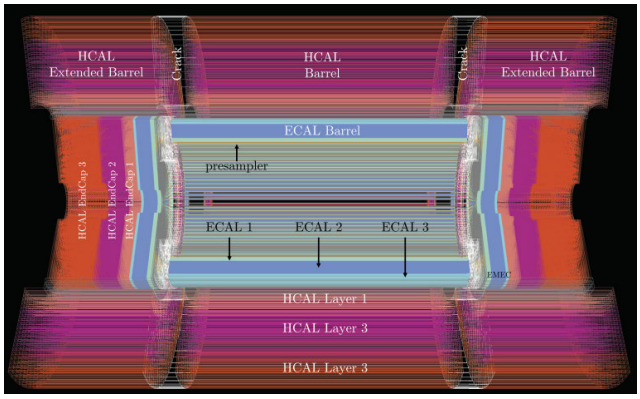


Fig. 1. Cross-sectional view of the general-purpose sampling calorimeter used for simulations with the Lorenzetti Showers framework. The single electron simulations were constrained to the barrel region for this work. Source: [15].

TABLE I

SEGMENTATION (LAYERS AND SAMPLINGS), COVERAGE IN $|\eta|$ AND CELL GRANULARITY USED IN THE SIMULATED CALORIMETER. ELECTROMAGNETIC AND HADRONIC CALORIMETERS ARE SEGMENTED INTO THREE LONGITUDINAL LAYERS AND PRECEDED BY A PS LAYER. SOURCE: [15]

Layer	Sampling	Coverage	Granularity ($\Delta\eta \times \Delta\phi$)
Presampler	Barrel	$0.00 < \|\eta\| < 1.58$	0.025×0.1
	End-Cap	$1.50 < \|\eta\| < 1.80$	0.025×0.1
Electromagnetic Calorimeter			
Layer 1	Barrel	$0.00 < \|\eta\| < 1.55$	0.003×0.1
		$1.37 < \|\eta\| < 1.80$	0.003×0.1
	End-Cap	$1.80 < \|\eta\| < 2.00$	0.025×0.1
		$2.00 < \|\eta\| < 2.37$	0.006×0.1
		$2.37 < \|\eta\| < 3.20$	0.1×0.1
Layer 2	Barrel	$0.00 < \|\eta\| < 1.50$	0.025×0.025
	End-Cap	$1.35 < \|\eta\| < 2.50$	0.025×0.025
		$2.50 < \|\eta\| < 3.20$	0.1×0.1
Layer 3	Barrel	$0.00 < \|\eta\| < 1.58$	0.05×0.1
	End-Cap	$1.35 < \|\eta\| < 2.50$	0.05×0.025
		$2.50 < \|\eta\| < 3.20$	0.1×0.1
Hadronic Calorimeter			
Layer 1	Barrel	$0.00 < \|\eta\| < 1.09$	0.1×0.1
	Extended Barrel	$0.94 < \|\eta\| < 1.77$	0.1×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2
Layer 2	Barrel	$0.00 < \|\eta\| < 1.09$	0.1×0.1
	Extended Barrel	$0.85 < \|\eta\| < 1.41$	0.1×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2
Layer 3	Barrel	$0.85 < \|\eta\| < 0.72$	0.2×0.1
	Extended Barrel	$0.85 < \|\eta\| < 1.41$	0.2×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2

noise follows a Gaussian distribution centered at zero, with standard deviation varying by calorimeter layer. Pileup noise is simulated by merging minimum-bias events into the primary event, as detailed in [15].

A cross-sectional view of the simulated calorimeter is shown in Fig. 1, and detailed segmentation parameters are provided in Table I.

To ensure comprehensive coverage of the calorimeter response under well-controlled conditions, single electrons in

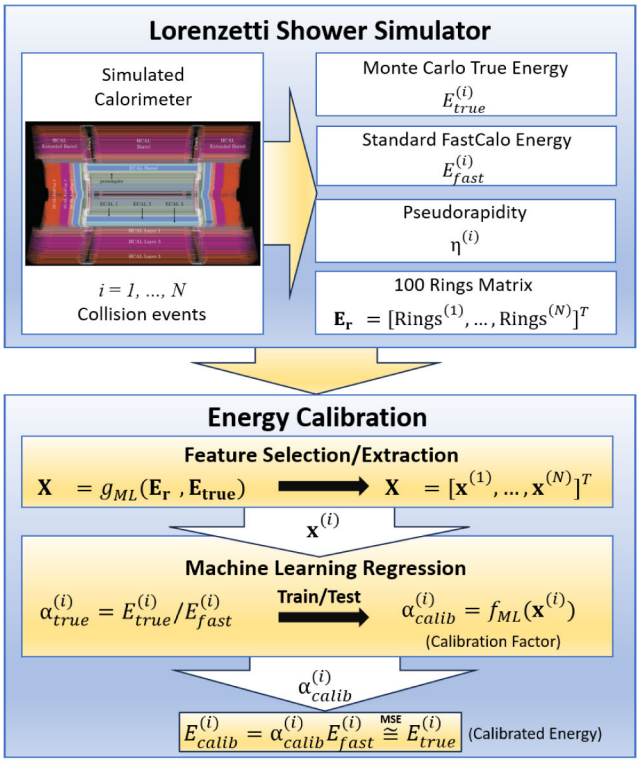


Fig. 2. Diagram of the proposed energy calibration system integrated with Lorenzetti Showers. The feature extraction and ML regression blocks are highlighted as the main processing steps for producing the calibration factors.

this study were generated using a dedicated *electron gun* configuration [23]. This setup enables the injection of electrons with predefined energies and incident angles directly into the detector volume, allowing systematic exploration across a wide kinematic range. The use of such generic, non-resonant electrons is particularly advantageous for calibration studies, as it supports the development of energy correction algorithms that remain valid over diverse conditions. In contrast, relying exclusively on decay-based electrons, such as those originating from Z boson decays, may restrict the calibration to narrow energy ranges and angular distributions. The particle-gun approach therefore ensures broader representativity and enhances the robustness of the resulting calibration models [10], [23], [24].

In each simulated event, a fixed-size region of interest (RoI) of 0.4×0.4 in (η, ϕ) is used to define the calorimetric region under analysis. When all seven layers of the Lorenzetti simulator are considered, this corresponds to approximately 1000 calorimeter cells per event. To reduce this high-dimensional input while preserving radial structure, the *Ringer* algorithm [8], [14] is applied within the fixed RoI. This algorithm integrates the energy deposits into 100 concentric rings centered on the shower axis, as illustrated in Fig. 2. These rings do not correspond to fixed geometrical areas in (η, ϕ). Instead, they are dynamically constructed based on radial distance from the shower barycenter (red square in Fig. 2), allowing the ring structure to adapt to the actual shape of each shower. This results in a 100-D feature vector that effectively captures the radial energy distribution and serves as input for the subsequent ML calibration models [17].

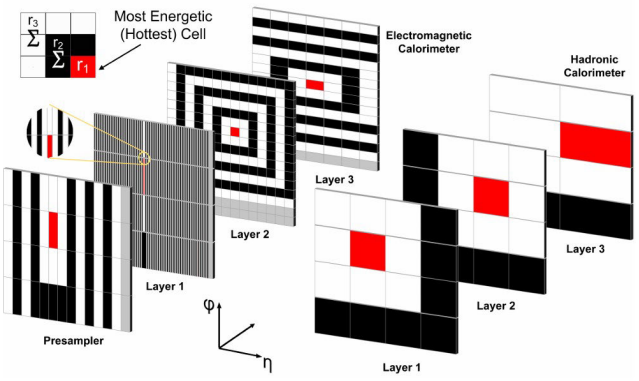


Fig. 3. Graphical representation of the ring-based reconstruction concept. The Ringer algorithm centers the reconstruction on the cell with the highest energy deposit (in red) and groups surrounding calorimeter cells into concentric rings across different calorimeter layers. Each ring aggregates the transverse energy deposited by the cells it encompasses, preserving the spatial topology of the electromagnetic shower in a 1-D profile. This structure serves as input to ML models such as the *Ringer* algorithm. Source: [25], [26].

Additionally, three physical quantities were recorded for each signature: the true energy (E_{true}), the standard trigger-level energy measurement (E_{fast}), and the pseudorapidity (η).

1) E_{true}*:* The true transverse energy of the particle, used as the reference for all other measurements. It represents a derived quantity obtained by projecting the particle's total energy onto the transverse plane with respect to the beam axis. In practice, it corresponds to the ideal transverse energy that would be measured if the particle's energy were deposited perpendicularly to the calorimeter surface, thus serving as the physical target for calibration.
2) E_{fast}*:* The standard FastCalo energy measurement refers to the uncalibrated energy values obtained from this system, which are primarily intended for rapid decision-making in the trigger chain. These measurements are commonly used as a baseline for comparison with more precise, calibrated energy estimates obtained offline. The term FastCalo originates from the fact that it represents a fast estimation of the energy deposited in the calorimeter, as provided by the trigger system.
3) η *(Pseudorapidity):* It describes the angle between the particle trajectory and the beam axis and is defined as $\eta = -\ln[\tan(\theta/2)]$, where θ is the polar angle with respect to the beam direction. In the context of this study, η corresponds to the pseudorapidity of the reconstructed electromagnetic object (e.g., electron) as it enters the virtual calorimeter simulated by the Lorenzetti Showers framework. This variable plays a critical role in defining binning strategies for the ML models and in evaluating performance across different detector regions, as will be discussed in the subsequent sections.

Fig. 3 illustrates the overall simulation and data processing workflow, highlighting the generation of derived variables and the sequential calibration stages leading to the estimation of the calibrated energy.

Section II-A details the physical modeling and mathematical formulation used to compute the fast transverse energy (E_{fast}) and to construct the 100 ring-based energy features employed by the *Ringer* algorithm for fast electron identification and calibration.

A. Standard FastCalo Energy and Ring Encoding

Starting from the raw calorimeter signals, energy is reconstructed at the cell level, aggregated into clusters, corrected for detector effects and pile-up, and finally transformed into structured ring features suitable for ML-based selection and energy calibration [17].

The energy in each calorimeter cell is estimated using optimal filtering of the digitized signal

$$E_k^{\text{cell}} = \sum_{j=1}^{N} w_j S_{kj} \tag{1}$$

where E_k^{cell} is the reconstructed energy in the kth cell, S_{kj} is the signal sampled at time j, and w_j is the optimal filter weight [17].

Reconstructed cell energies are aggregated within an adaptively defined shower cluster volume to compute the raw cluster energy

$$E_{\text{raw}} = \sum_{k\in\text{cluster}} E_k^{\text{cell}}. \tag{2}$$

The transverse component of the fast energy, or Standard FastCalo Energy, is computed using the cluster pseudorapidity

$$E_{\text{fast}} = \frac{E_{\text{raw}}}{\cosh(\eta)} \tag{3}$$

where η is the pseudorapidity of the cluster centroid, which determines the angle of incidence, as previously defined.

Additionally, the Ringer algorithm aggregates the energy into concentric rings centered on the electromagnetic shower core, as illustrated in Fig. 2 and formalized in the following equation:

$$E_{r,j} = \sum_{k\in\text{ring}\ j} \frac{E_k^{cell}}{\cosh(\eta)} \tag{4}$$

where $E_{r,j}$ is the total transverse energy in the jth ring and E_k^{cell} is the energy deposited in the kth calorimeter cell belonging to that ring, as shown in (1).

The ring energies are normalized to form a dimensionless energy profile

$$\tilde{E}_{r,j} = \frac{E_{r,j}}{\sum_{k=1}^{100} E_{r,k}} \tag{5}$$

where $\tilde{E}_{r,j}$ represents the normalized energy of the j-th ring. This normalization emphasizes the shower shape rather than its magnitude, making the resulting profile invariant to the absolute energy scale.

The complete matrix of normalized ring energies is structured as follows:

$$\tilde{\mathbf{E}}_r = \begin{bmatrix} \tilde{E}_{r,1}^{(1)} & \cdots & \tilde{E}_{r,100}^{(1)} \\ \tilde{E}_{r,1}^{(2)} & \cdots & \tilde{E}_{r,100}^{(2)} \\ \vdots & \ddots & \vdots \\ \tilde{E}_{r,1}^{(N)} & \cdots & \tilde{E}_{r,100}^{(N)} \end{bmatrix} = \begin{bmatrix} \text{Rings}\,[1{:}100]^{(1)} \\ \text{Rings}\,[1{:}100]^{(2)} \\ \vdots \\ \text{Rings}\,[1{:}100]^{(N)} \end{bmatrix} \tag{6}$$

where each row corresponds to a collision event and encodes the normalized energy profile across all 100 concentric rings.

From this point forward, for notational convenience, we define

$$\text{Rings}\,[1{:}100]^{(i)} = \left[\tilde{E}^{(i)}_{r,1}, \ldots, \tilde{E}^{(i)}_{r,100}\right] \tag{7}$$

to represent the full set of 100 rings for the ith event and $\text{Rings}[j]^{(i)}$ as the energy of the jth ring for the same event.

Initial measurements of E_{fast} serve as the first estimation of transverse energy in this process. However, due to limitations in resolution and detector effects, these values require refinement to achieve higher accuracy. The objective is to calibrate E_{fast} such that the resulting calibrated energy E_{calib} better approximates the true transverse energy E_{true} while minimizing the associated variance.

This motivates the development of a dedicated calibration strategy, detailed in the following subsections, which proposes a data-driven approach to estimate a correction factor for each event. The method leverages energy deposition patterns extracted from calorimeter rings and applies ML techniques to minimize the discrepancy between E_{calib} and E_{true} in the mean squared error (mse) sense.

B. Calibration Method

The energy calibration method proposed in this work is based on estimating a calibration factor $\alpha^{(i)}_{\text{calib}}$ to adjust the reconstructed transverse energy $E^{(i)}_{\text{fast}}$, such that the calibrated energy $E^{(i)}_{\text{calib}}$ approaches the true transverse energy $E^{(i)}_{\text{true}}$, for the i-th collision event. The method involves dimensionality reduction of the energy ring inputs and estimation of the calibration factor using an ML model. The procedure can be summarized by the following steps.

1) Definition of calibrated transverse energy

$$E^{(i)}_{\text{calib}} = \alpha^{(i)}_{\text{calib}} \cdot E^{(i)}_{\text{fast}}. \tag{8}$$

2) Calculation of the targeted true calibration factor

$$\alpha^{(i)}_{\text{true}} = \frac{E^{(i)}_{\text{true}}}{E^{(i)}_{\text{fast}}}. \tag{9}$$

3) Dimensionality reduction of energy rings

$$\mathbf{X} = g_{\text{ML}}\left(\tilde{\mathbf{E}}_r, \mathbf{E}_{\text{true}}\right). \tag{10}$$

4) Training ML models to estimate calibration factor

$$\alpha^{(i)}_{\text{calib}} = f_{\text{ML}}\left(\mathbf{x}^{(i)}\right). \tag{11}$$

5) Estimation of calibrated transverse energy

$$E^{(i)}_{\text{calib}} = f_{\text{ML}}\left(\mathbf{x}^{(i)}\right) \cdot E^{(i)}_{\text{fast}} \approx E^{(i)}_{\text{true}}. \tag{12}$$

6) Performance evaluation with mse loss function

$$\text{mse}\,(E) = \frac{1}{N}\sum_{i=1}^{N}\left(E^{(i)}_{\text{true}} - f_{\text{ML}}\left(\mathbf{x}^{(i)}\right) \cdot E^{(i)}_{\text{fast}}\right)^2 \tag{13}$$

where $i = 1, 2, \ldots, N$, an N is the total number of collision events

$$\mathbf{X} = \left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(N)}\right]^T \tag{14}$$

where $\mathbf{x}^{(i)} \in \mathbb{R}^{1\times F}$ is a vector with F features per event i after dimensionality reduction.

This formulation encapsulates the full ML-based calibration method, where $g_{\text{ML}}(\cdot)$ performs feature selection or extraction from the energy ring matrix ($\tilde{\mathbf{E}}_r \in \mathbb{R}^{N\times 100} \rightarrow \mathbf{X} \in \mathbb{R}^{N\times F}$, with $F \ll 100$), and $f_{\text{ML}}(\cdot)$ predicts the event-wise correction factor $\alpha^{(i)}_{\text{calib}}$. The goal is to minimize the mse mse(E), ensuring that the calibrated energy closely approximates the true transverse energy, represented by the vector $\mathbf{E}_{\text{true}} = [E^{(1)}_{\text{true}}, \ldots, E^{(N)}_{\text{true}}]^T$. Here, $\mathbf{x}^{(i)} \in \mathbb{R}^{1\times F}$ denotes the reduced feature vector for event i, obtained after dimensionality reduction to F attributes, with $F \ll 100$.

In the following, we detail the construction of these reduced feature vectors.

C. Features Selection and Extraction

For feature selection, we employed correlation (CORR) and mutual information (MUTI) to identify attributes from the energy rings that have the highest relevance for estimating E_{true}. Additionally, we have used multivariate feature extraction methods such as principal component analysis (PCA) and independent component analysis (ICA) [27], [28], [29].

The input matrix for all dimensionality reduction techniques is defined as $\mathbf{X} = g_{\text{ML}}(\tilde{\mathbf{E}}_r, \mathbf{E}_{\text{true}}) \in \mathbb{R}^{N\times F}$, where each row corresponds to a collision event, and each column represents a selected or transformed feature extracted from the normalized ring energies. We denote by $\mathbf{X}^{(j)} \in \mathbb{R}^N$ the j-th column of $\mathbf{X}$, which contains the values of the j-th feature across all N events. We evaluate four different methods for constructing $\mathbf{X}$, each applying a specific criterion to identify the most informative features:

1) *Correlation (CORR):* Features $\mathbf{X}^{(j)}$, for $j = 1, \ldots, 100$, are selected such that

$$|\rho\left(\mathbf{X}^{(j)}, \mathbf{E}_{\text{true}}\right)| \geq \tau \tag{15}$$

where $\rho(\cdot,\cdot)$ is the Pearson correlation coefficient defined as

$$\rho\,(\mathbf{u}, \mathbf{v}) = \frac{\sum_{i=1}^{N}(u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^{N}(u_i - \bar{u})^2}\sqrt{\sum_{i=1}^{N}(v_i - \bar{v})^2}} \tag{16}$$

with $\bar{u}$ and $\bar{v}$ denoting the sample means of vectors $\mathbf{u}$ and $\mathbf{v}$, respectively [30]. Threshold values $\tau \in \{0.50, 0.70, 0.80, 0.90\}$ are tested to evaluate the impact of filtering by correlation strength.

2) *Mutual Information (MUTI):* Features are selected according to

$$\text{MI}\left(\mathbf{X}^{(j)}, \mathbf{E}_{\text{true}}\right) \geq \tau \tag{17}$$

where $\text{MI}(\cdot,\cdot)$ is the mutual information defined as

$$\text{MI}\,(U, V) = \sum_{u\in\mathcal{U}}\sum_{v\in\mathcal{V}} p\,(u, v)\log\left(\frac{p\,(u, v)}{p\,(u)\,p\,(v)}\right) \tag{18}$$

with $p(u, v)$ representing the joint probability distribution of U and V, and $p(u), p(v)$ their marginals [31]. This method captures both linear and nonlinear dependencies.

3) *PCA:* It transforms the input matrix $\tilde{\mathbf{E}}_r \in \mathbb{R}^{N\times 100}$ into an orthogonal basis of uncorrelated components

$\mathbf{PCA}^{(j)}$, ordered by the variance they explain. To ensure relevance to the target, components are sorted by absolute Pearson correlation with $\mathbf{E}_{\text{true}}$, and the top $m \in \{100, 10, 5, 2\}$ components are selected

$$|\rho\left(\mathbf{PCA}^{(1)}, \mathbf{E}_{\text{true}}\right)| \geq \cdots \geq |\rho\left(\mathbf{PCA}^{(m)}, \mathbf{E}_{\text{true}}\right)|. \quad (19)$$

4) *ICA:* It decomposes the ring energy matrix into statistically independent components $\mathbf{ICA}^{(j)}$, which are then sorted by their absolute correlation with the target

$$|\rho\left(\mathbf{ICA}^{(1)}, \mathbf{E}_{\text{true}}\right)| \geq \cdots \geq |\rho\left(\mathbf{ICA}^{(m)}, \mathbf{E}_{\text{true}}\right)|. \quad (20)$$

The top $m \in \{100, 10, 5, 2\}$ components are selected to form the reduced matrix $\mathbf{X} \in \mathbb{R}^{N \times m}$. This method is especially useful when the informative structure is based on latent, non-Gaussian independent sources [29].

To simplify the presentation of results, we adopt the following notation for each method: CORR-τ indicates the correlation-based selection with threshold τ (e.g., CORR-70 selects features with Pearson correlation above 0.70); MUTI-τ refers to mutual information selection with threshold τ (e.g., MUTI-80 retains features with mutual information above 0.80); PCA-m denotes PCA with the top m components retained (e.g., PCA-2 uses only the two most informative principal components); and ICA-m indicates ICA with the top m components selected (e.g., ICA-10 retains the ten most relevant independent components).

In addition to computing the Pearson correlation coefficients between each extracted component and the target variable $\mathbf{E}_{\text{true}}$, we also evaluated the proportion of explained variance associated with each principal component (in the case of PCA) and each independent component (in the case of ICA). For PCA, this is derived from the eigenvalues of the correlation matrix, quantifying the amount of total variance retained by each component. For ICA, although the components are not ranked by variance by default, we apply a post-hoc analysis of the variance explained to assess their relative informativeness.

This analysis provides insight into how much of the original dataset's variability is captured by the selected components, supporting the interpretability and effectiveness of the dimensionality reduction process.

Upon completion of the feature selection or extraction step, the resulting matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$ contains the reduced set of attributes used as input to the ML models described in the following.

D. ML for Regression

Regression methods aim to learn a mapping $\alpha_{\text{calib}}^{(i)} = f_{\text{ML}}(\mathbf{x}^{(i)})$ from the reduced feature vector $\mathbf{x}^{(i)} \in \mathbb{R}^F$ to the calibration factor $\alpha_{\text{calib}}^{(i)}$. The choice of regression algorithm directly impacts the accuracy, generalization, and computational efficiency of the calibration procedure. Each method presents distinct capabilities and trade-offs, enabling a comprehensive evaluation of its suitability for energy calibration. The approaches considered are as follows [28].

1) *Multivariate Linear Regression Without $|\eta| \times E_T$ Binning (MLR no-Bin):* MLR assumes a linear dependency between features and the output variable α_{calib}. It is computationally efficient and interpretable but limited in modeling nonlinear behaviors, often leading to suboptimal calibration accuracy and serving as a baseline reference.
2) *Polynomial Regression of Degree 2 Without $|\eta| \times E_T$ Binning (PREG no-Bin):* PREG extends MLR by incorporating second-order terms, enabling the capture of basic nonlinearities. While offering improved flexibility, it remains susceptible to overfitting, particularly in high-dimensional or noisy datasets.
3) *Decision Trees Without $|\eta| \times E_T$ Binning (DTREE no-Bin):* DTREE models nonlinear interactions by recursively partitioning the feature space using hierarchical decision rules. It is capable of capturing complex patterns but can easily overfit unless properly regularized.
4) *Gradient BDTs (BDT no-Bin):* This approach employs an ensemble of shallow trees trained iteratively to minimize prediction error. BDTs are well-suited for capturing subtle interactions and noise patterns in the data, yielding robust performance without requiring binning [32].
5) *Gradient BDTs With $|\eta| \times E_T$ Binning (BDT Bin):* This strategy first segments the dataset into bins defined over pseudorapidity $|\eta|$ and transverse energy E_T and then trains independent BDTs per bin. Binning allows the models to specialize in region-specific detector responses, improving local calibration precision. However, small bin sizes may reduce statistical robustness in sparse regions [11].

The selection of regression methods in this study was guided not only by their modeling capabilities but also by their computational efficiency, with a view toward real-time applicability. Since one of the intended use cases is the integration of these calibration models into an online trigger system, it is essential to prioritize algorithms that offer a favorable trade-off between accuracy and processing time. Trigger-level applications in HEP impose strict latency and resource constraints, especially at the high-level trigger (HLT) stage. Therefore, we initially focus on evaluating methods that are lightweight in terms of inference cost, such as linear and shallow decision tree-based approaches, before considering more complex architectures. This strategy ensures that the calibration models remain feasible for deployment in operational environments with stringent timing requirements.

In addition to evaluating different regression algorithms, we also compare two training strategies: using the entire dataset without binning and segmenting the data by $|\eta| \times E_T$. The segmentation approach requires defining suitable binning boundaries based on the Lorenzetti Shower simulation, enabling the models to specialize in distinct detector response regions.

Segmenting the dataset by pseudorapidity $|\eta|$ and transverse energy E_T is a widely adopted practice in HEP ML workflows [13], [33], as it allows the models to better capture spatial and energy-dependent non-uniformities in the calorimeter. Although this strategy can enhance local model accuracy, it comes at the cost of splitting the training data, which may lead to insufficient statistics in certain bins and impair generalization.

To address this limitation, we applied a genetic algorithm (GA) to optimize the binning scheme over the $|\eta| \times E_T$ space. The optimization objective was to ensure a statistically balanced distribution of events across bins while maintaining high calibration performance. This approach guarantees that each specialized model is trained with a sufficient number of events to preserve both statistical robustness and generalization capacity.

E. GA for Binning

Each individual (chromosome) in the population encodes a candidate segmentation, represented by two ordered vectors of real values: one for the E_T bin boundaries and another for the $|\eta|$ bin boundaries. In this study, the number of bins was fixed at 5 for E_T and 3 for $|\eta|$, resulting in a total of 15 segments. The vectors define flexible, nonuniform boundaries between 0 and the respective maximum values observed in the dataset.

The GA was configured with a population size of 100, a mutation rate of 20%, and a maximum of 200 generations. The fitness function evaluates the statistical uniformity of the segments, minimizing the difference between the largest and smallest bin populations. The initial population is generated through random perturbations of evenly spaced bins, and the evolution process includes elitism, crossover, and multiple mutation strategies to preserve diversity. Early stopping is triggered if no improvement in the population's average fitness is observed over 30 generations. The best-performing chromosome defines the final segmentation configuration used in the subsequent calibration.

This approach maintains a relevant number of data points in each bin, preventing sparsity and enabling robust energy estimation. Table II illustrate the final binning for $|\eta|$ and E_t.

1) $|\eta| = [0, 0.43, 0.83, 1.46]$.
2) $E_t = [0, 44.7, 71.5, 103.5, 164.3, 499.9]$ (GeV).

Now that the ML regression methods have been introduced and the binning strategy has been defined using a GA, we proceed to evaluate the performance of each configuration. The combination of the four feature selection or extraction methods (CORR, MUTI, PCA, and ICA), each tested with four thresholds or component counts, with the five ML regression models, results in a total of 80 distinct scenarios. These scenarios are systematically analyzed to identify the most effective combination for accurate energy calibration.

To determine the best-performing configurations among these 80 scenarios, we rely on a rigorous error evaluation framework, described in detail in the next section.

F. Energy Error Evaluation

The objective here is to quantitatively compare the performance of different calibration methods by analyzing the discrepancies between the calibrated energy values E_{calib} and the true energy E_{true}. The calibration error for each event is defined as

$$\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}} \tag{21}$$

where calib refers to the ML model used for estimating the calibration factor. For the baseline reference, the error from the standard method (FastCalo) is computed as $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$.

An ideal estimator would yield zero mean error ($\mu_\varepsilon = 0$) and zero error variance ($\sigma_\varepsilon^2 = 0$). To evaluate both the accuracy and the stability of the calibration, two complementary performance metrics are adopted.

1) RMSE measures the average magnitude of the error, giving higher weight to large deviations.
2) IQR quantifies the dispersion of the error around the median, being robust to outliers and non-Gaussian tails.

RMSE is particularly informative for assessing the overall calibration accuracy, while IQR captures the consistency of the predictions across the dataset. These metrics were selected for their interpretability, low computational cost, and direct relevance in HEP applications, where both precision and robustness are critical [28]. In this work, we therefore focused on mean residuals, variance ratios, RMSE, and IQR, complemented by rigorous hypothesis testing, to ensure practical applicability in real-time trigger environments. Although more complex alternatives, such as divergence-based measures (e.g., Kullback–Leibler divergence or Wasserstein distance), could provide complementary insights, they were not adopted here in order to preserve interpretability and computational efficiency. Such divergence metrics remain a promising direction for future studies, particularly in analyses with real experimental data [28].

To consolidate both metrics into a single figure of merit, we compute the Euclidean distance from the origin in the RMSE–IQR plane

$$d_0 = \sqrt{\text{RMSE}^2 + \text{IQR}^2}. \tag{22}$$

This scalar score summarizes the trade-off between accuracy and variability. It is important to highlight that d_0 is focused solely on predictive accuracy and precision; it does not incorporate factors such as computational cost, model complexity, or memory footprint, which may also be relevant in large-scale deployment scenarios.

Therefore, to complement the error-based evaluation and provide a more comprehensive analysis of each method's practical viability, we now assess their computational efficiency in terms of feature dimensionality and inference latency.

G. Number of Features and Processing Time Evaluation

In addition to predictive accuracy, the computational efficiency of each method is a critical factor for its applicability in high-throughput environments such as real-time trigger systems in HEP experiments. This section evaluates each configuration based on two main criteria: the number of features used during inference and the prediction time per event.

1) *Number of Features (F):* A lower number of features generally leads to reduced memory consumption, lower computational complexity, and faster processing. This metric is directly determined by the dimensionality reduction strategy applied to the normalized ring energy matrix.
2) *Processing Time (T):* Defined as the inference latency per event, this metric reflects the time required by the trained model to predict the calibration factor. While

training time was monitored, it is excluded from this evaluation since model training is performed offline and does not impact real-time performance.

For benchmarking purposes, the Standard FastCalo energy is used as a reference baseline, with only one input feature ($F_{\text{fast}} = 1$) and effectively zero inference latency ($T_{\text{fast}} = 0$), as its output is directly computed from simulation parameters, without requiring additional processing.

To consolidate the evaluation of predictive performance and computational cost, we define a global score D_0 that integrates four metrics, IQR, RMSE, number of features (F), and inference time (T), into a single normalized scalar

$$D_0 = \sqrt{IQR_{(N)}^2 + RMSE_{(N)}^2 + F_{(N)}^2 + T_{(N)}^2} \tag{23}$$

where $IQR_{(N)}$, $RMSE_{(N)}$, $F_{(N)}$, and $T_{(N)}$ denote the min-max normalized values of each metric across all 80 configurations. Normalization is essential due to the heterogeneous scales and units of the metrics, ensuring that the components are directly comparable before aggregation. In the definition of D_0, equal weights were deliberately assigned to all four terms. This choice reflects the intent of D_0 as a baseline, non-biased aggregator, avoiding the arbitrariness that would result from subjectively prioritizing predictive accuracy (RMSE and IQR) over computational efficiency (inference time and number of features) or vice versa. Since both aspects are critical in real-time trigger applications, equal weighting provides a transparent and reproducible compromise between accuracy and efficiency.

Furthermore, to assess robustness, we also reported results using the d_0 score, which aggregates only RMSE and IQR. The high consistency between the top-ranked methods according to d_0 and D_0 indicates that the ranking is not overly sensitive to the exact weighting choice. The observed differences between the two scores mainly arise from efficiency-related gains, as expected by construction. Thus, lower values of D_0 indicate better overall trade-offs, while the agreement with d_0 supports the stability of the ranking across different weighting schemes.

Inference times were measured using the following hardware configuration.

1) *Processor:* Intel[1] Core[2] i7-7500U.
2) *RAM:* 20 GB DDR4.
3) *Storage:* 1 TB SSD.
4) *GPU:* NVIDIA GeForce1 940MX (2 GB).

While modest compared to the infrastructure used in large-scale experiments HLT systems [17], [34], [35], this setup offers a conservative and robust basis for testing the practical feasibility of the proposed models. Given that latency thresholds in this evaluation align with those required for real-time operations, the ability of a method to meet them on this hardware implies potential deployability in high-performance environments.

Having ranked the configurations by their global D_0 scores, we now proceed to a more granular evaluation of the top-performing models. The next section investigates how the most promising methods perform within each $|\eta| \times E_t$ bin defined earlier, focusing on the stability and accuracy of the estimated calibration factors.

[1]Registered trademark.
[2]Trademarked.

TABLE II
COUNT OF EVENTS PER BIN COMBINATION AFTER GA OPTIMIZATION

E_t interval (GeV)	$\lvert\eta\rvert \in [0, 0.43]$	$[0.43, 0.83]$	$[0.83, 1.46]$
$[0, 44.7]$	5462	6773	15347
$[44.7, 71.5]$	6310	7193	12677
$[71.5, 103.5]$	8160	8927	14772
$[103.5, 164.3]$	15305	15642	7955
$[164.3, 499.9]$	13493	7769	5954

H. Calibration Factor per Bin Evaluation

Following the global ranking based on the D_0 metric, the top four configurations were selected for a more detailed, localized analysis within each $|\eta| \times E_t$ bin, as defined in Table II. This bin-wise evaluation aims to assess the consistency and stability of the predicted calibration factor α_{calib} across the detector acceptance.

The analysis considers three complementary metrics: IQR, mean absolute error (MAE), and RMSE, all computed with respect to the true calibration factor. The ground-truth calibration target used during training is defined in (9), which represents the optimal multiplicative correction to be applied to the FastCalo energy estimate E_{fast} in order to recover the true transverse energy E_{true}.

The bin-wise IQR of α_{calib} serves as a robust measure of statistical dispersion, indicating the variability of predictions around the median within each region. Meanwhile, MAE and RMSE quantify the absolute and squared deviations of the predicted calibration factor from the ideal correction α_{true}, providing insights into both accuracy and sensitivity to outliers.

This localized analysis is crucial for identifying potential instabilities or systematic deviations in specific detector regions that may not be evident in global performance metrics. However, to ensure that the observed differences between methods, both globally and per bin, are statistically robust and not the result of random fluctuations, a formal statistical validation is required.

I. Statistical Validation

To rigorously determine whether the observed improvements of the calibrated method over the standard FastCalo approach are genuine and not attributable to random variation, we performed a comprehensive statistical analysis combining parametric and non-parametric one-sided paired hypothesis tests [36], [37], [38], [39], [40]. The paired design reflects the fact that each calibrated and uncalibrated measurement originates from the same physical event, preserving within-event correlation and increasing the power of the tests. An initial assessment of normality for the error distributions and their paired differences guided the choice of methods; large-sample parametric tests were applied as primary tools, supported by robust non-parametric alternatives to ensure validity under deviations from Gaussianity. All tests were carried out at a significance level of $\alpha = 0.05$ under the directional alternative hypothesis that the calibrated method yields systematically lower errors—both in mean and variance—than the baseline.

1) Normality Assessment: Before selecting the appropriate paired tests, we evaluated the normality of the calibration errors $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$ and $\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$, and their paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$. We applied the D'Agostino–Pearson K^2 test [41], [42] and the Anderson–Darling test [43], [44], [45], both of which are well-suited for large samples and have good sensitivity to tail departures from normality. The Shapiro–Wilk test was omitted for $n > 5000$ due to known computational and interpretative limitations in very large datasets [44].

Normality diagnostics also included visual inspection via quantile–quantile (Q–Q) plots, which provide qualitative confirmation of distributional deviations and highlight heavy tails and asymmetry [46]. Although rejection of normality was anticipated due to the large sample sizes, these tests guided the decision to complement parametric inference with robust non-parametric alternatives.

2) Tests on the Mean Error: The primary objective was to test

$$H_0 : \mu_d \geq 0 \quad \text{vs.} \quad H_1 : \mu_d < 0$$

where μ_d is the mean of the paired differences d. The following methods were applied.

1) *Large-Sample Paired z-Test (Primary Method):* Designed for *paired* observations, this test is applied to the differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$. For large sample sizes, the central limit theorem ensures that the sampling distribution of the mean $\bar{d}$ is approximately normal regardless of the distribution of d [36], [47], [48]. The standard error of $\bar{d}$ is estimated from the sample, and inference is made via a one-sided 95% confidence interval (CI) for μ_d, which is mathematically equivalent to the hypothesis test. The CI is therefore not a confirmatory second step, but an alternative expression of the same decision rule [49]. This test is robust to non-normality in large-n paired settings but can be complemented by non-parametric methods when strong deviations from symmetry or heavy tails are present [48], [50].
2) *Paired Student's t-Test (Reference):* The exact small-sample parametric test for μ_d under the assumption that d is normally distributed with unknown variance [38], [39]. It accounts for the correlation induced by pairing by operating directly on the within-pair differences. As n increases, the t statistic converges to the z statistic, making both approaches equivalent in large samples [36], [38]. When normality is violated in small samples, the test's validity can be compromised, motivating the parallel use of robust alternatives [48], [50].
3) *Wilcoxon Signed-Rank Test (Robustness Check):* A non-parametric procedure specifically designed for *paired* data, testing whether the median of the differences d is zero [50], [51]. It makes no distributional assumption beyond the symmetry of d around its median and is resistant to the influence of outliers, skewness, and heavy-tailed distributions [39]. This test provides a robustness check against violations of normality, complementing the parametric t and z tests [50].

3) Tests on the Error Variance: We also tested

$$H_0 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} \geq 1 \quad \text{vs.} \quad H_1 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} < 1$$

where σ^2_{calib} and σ^2_{fast} are the variances of $\varepsilon_{\text{calib}}$ and $\varepsilon_{\text{fast}}$, respectively. The following methods were used.

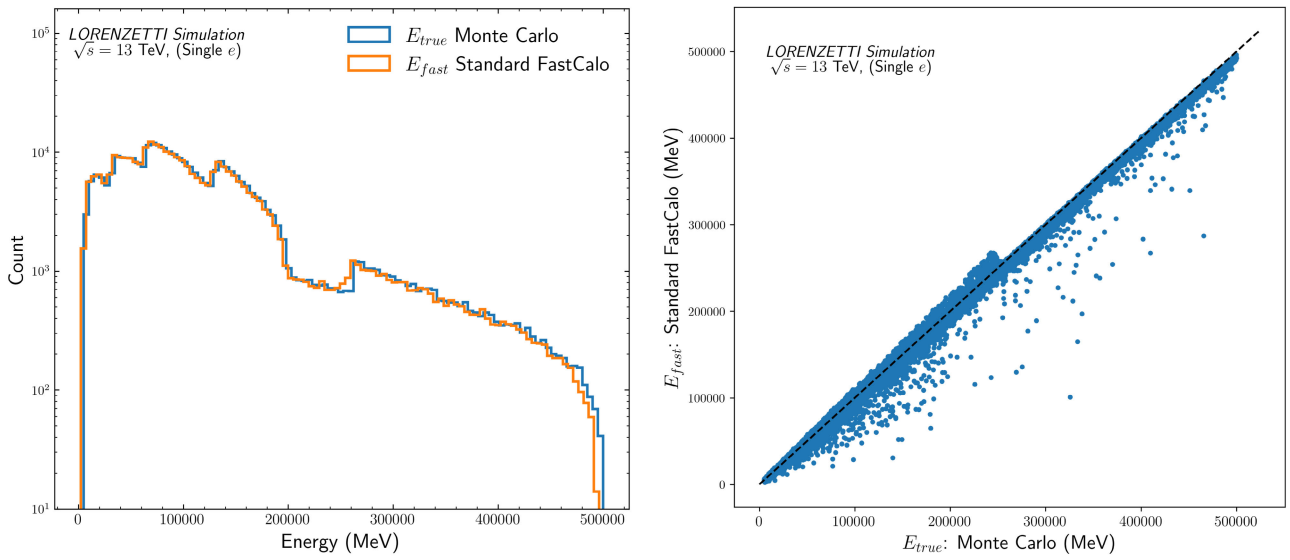
1) *Pitman–Morgan Test (Primary for Variance):* Specifically developed for *paired* samples, this test evaluates the equality of variances while explicitly accounting for the correlation between paired observations [52], [53]. Under the assumption of bivariate normality of $(\varepsilon_{\text{calib}}, \varepsilon_{\text{fast}})$, the test is generally more powerful than independent-sample variance tests. In the presence of strong deviations from normality (e.g., heavy tails), it can exhibit inflated Type I error, so it was complemented by robust methods [38].
2) *Paired Bootstrap BCa Test (Robustness Check):* A non-parametric resampling approach applied directly to the variance ratio $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$. Although the log transformation is sometimes employed to improve symmetry, in this analysis, the ratio was used without transformation to retain interpretability in its natural units. The bias-corrected and accelerated (BCa) CIs provide accurate coverage even when the sampling distribution of the ratio is skewed or when the normality assumption is violated. This method preserves the pairing structure by resampling event pairs jointly [54], [55].
3) *F-Test for Variances (Comparative Reference):* The classical F-test for variance ratios [36], [37] assumes independent samples from normal populations and does not incorporate the correlation present in paired designs. It is included here solely for comparability with conventional results in the literature; in the presence of paired data or nonnormal residuals, its Type I error rate can be substantially affected.

For each method, we report the test statistic, p-value, and the corresponding one-sided 95% CI for the effect size (mean difference in MeV or variance ratio, dimensionless). A CI entirely below zero (for μ_d) or below one (for variance ratios) is equivalent to rejecting H_0 at $\alpha = 0.05$.

III. Results

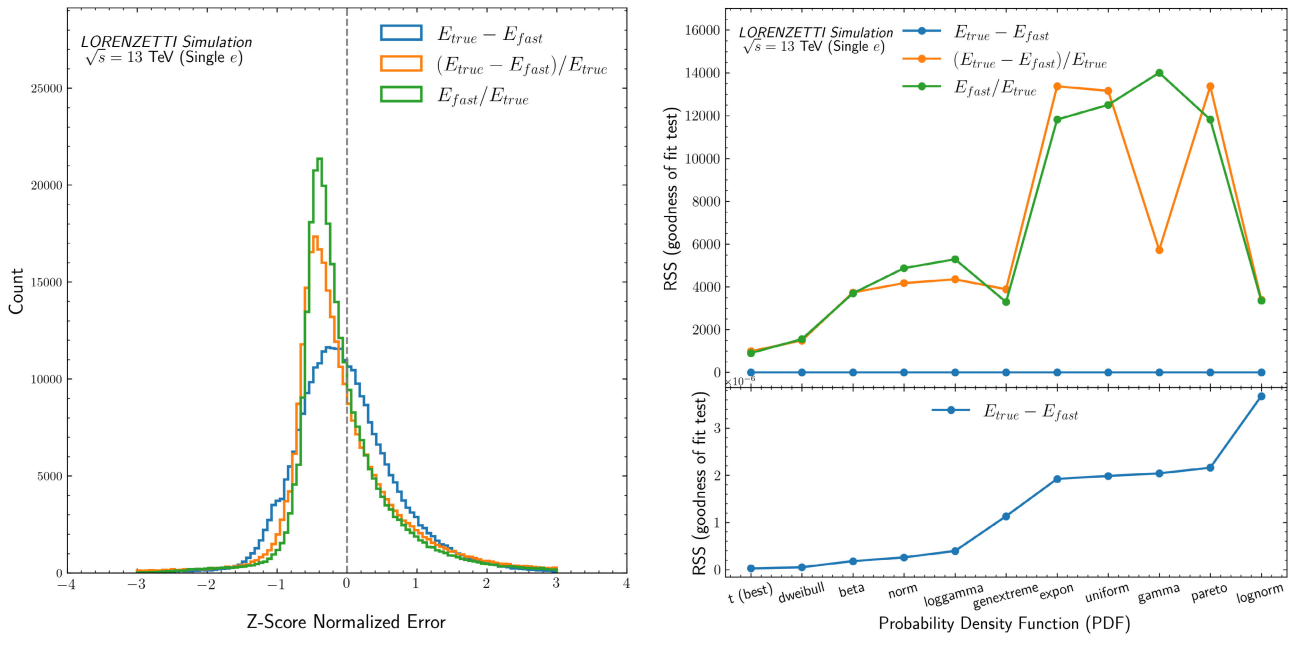
Fig. 4(a) presents the histograms of True Energy (E_{true}) and Standard FastCalo Energy (E_{fast}), both showing similar distributions. However, discrepancies are evident in the scatter plot depicted in Fig. 4(b), where distortions in mean and variance are apparent. Specifically, mean distortions exhibit a discontinuity around 250 GeV, while variance distortions are asymmetric, with greater dispersion observed below the $E_{\text{true}} = E_{\text{fast}}$ axis.

Further analysis in Fig. 5(a) examines the relationships between E_{true} and E_{fast}, including the calibration error $E_{\text{true}} - E_{\text{fast}}$, normalized calibration error $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$, and calibration factor $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$. All plots are normalized using the z-score for better comparison. Z-score is defined as the number of standard deviations a data point is from the mean [36].



(a) (b)

Fig. 4. Comparison of true and fast reconstructed energies. (a) Energy histogram comparing true energy (E_{true}) and standard fast reconstructed energy (E_{fast}) using the Lorenzetti simulator, over the 0–500 GeV range, revealing minor discrepancies between the two measurements. (b) Scatter plot of true energy (E_{true}) versus standard fast measured energy (E_{fast}), showing overall correlation but with noticeable nonlinear distortions in the mean and variance above 250 GeV.



(a) (b)

Fig. 5. Comparison of energy errors and goodness of fit tests of errors. (a) Z-score normalized relationships between E_{true} and E_{fast} illustrating energy errors. The legend presents different methods of evaluating the error: the absolute difference ($E_{\text{true}} - E_{\text{fast}}$), the relative difference normalized by the true energy ($E_{\text{true}} - E_{\text{fast}}$)/$E_{\text{true}}$, and the ratio $E_{\text{fast}}/E_{\text{true}}$. The objective is to identify the relation whose distribution is closest to a Gaussian, thereby simplifying the treatment of calibration errors. (b) Residual sum of squares (RSS) goodness-of-fit tests were performed to evaluate different formulations of the calibration error between E_{true} and E_{fast}, aiming to determine the most statistically appropriate distribution for each case. Among the evaluated formulations, the absolute error ($E_{\text{true}} - E_{\text{fast}}$) yielded the lowest RSS, indicating the best overall fit. The blue curve, corresponding to this formulation, presented significantly smaller residuals compared to the green and orange curves, appearing nearly flat in the top plot. A magnified view of the blue curve is shown in the bottom plot. The results suggest that the Student's t-distribution provides the best fit for modeling the calibration error.

Fig. 5(b) shows that the residual distribution $E_{\text{true}} - E_{\text{fast}}$ closely follows a Student's t-distribution, in contrast to the more skewed behaviors observed for the relative errors ($E_{\text{true}} - E_{\text{fast}}$)/$E_{\text{true}}$ and the energy ratio $E_{\text{true}}/E_{\text{fast}}$, both of which deviate significantly from normality. These results indicate that the absolute error $E_{\text{true}} - E_{\text{fast}}$ is a more appropriate metric for statistical inference. Consequently, it was adopted in the t and F-tests used for the validation of calibration accuracy and variance.

The calibration factor α_{true}, crucial for ML methods estimating α_{calib}, is examined in Fig. 6. The boxplots across different E_t and $|\eta|$ bins reveal that estimated calibration factors should ideally match α_{true} to minimize calibration errors ($\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$).

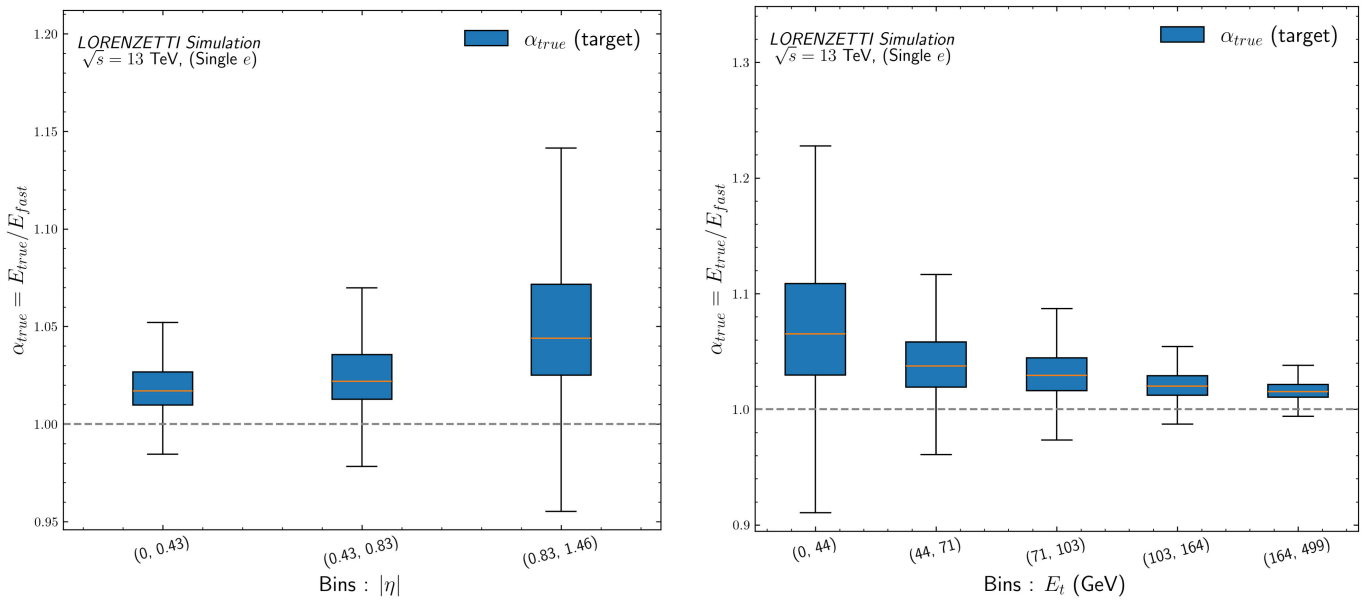


Fig. 6. Boxplots of the true calibration factor α_{true} across (a) pseudorapidity $|\eta|$ and (b) transverse energy E_{t} bins. The boxplots illustrate the distribution of calibration factors within each bin, where the central line represents the median, the box denotes the IQR, and the whiskers extend to 1.5 times the IQR from the lower and upper quartiles. This visualization highlights the uncertainty and variability of the calibration factors per bin, which serve as the target for the ML calibration methods.

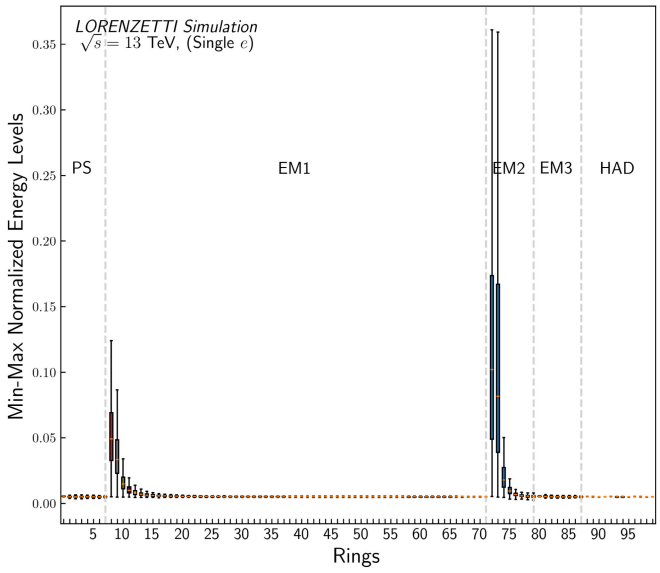


Fig. 7. Boxplot of min-max normalized energy levels across the 100 concentric rings defined by the *Ringer* algorithm for 300 000 simulated single-electron events. Each box represents the distribution of normalized energy values in one ring across all events. Vertical dashed lines delineate the approximate positions of the calorimeter layers: PS, electromagnetic layers (EM1–EM3), and Hadronic layers (HAD). The concentration of energy in EM2 is consistent with typical electromagnetic shower development.

Fig. 7 presents the distribution of normalized energy depositions across the 100 concentric rings generated by the *Ringer* algorithm for the full set of 300 000 simulated events. Each box represents the IQR of the normalized energy levels in a given ring, obtained via min-max normalization applied independently to each event.

The vertical dashed lines indicate the approximate boundaries corresponding to the different longitudinal layers of the calorimeter: the pre-sampler (PS), the electromagnetic layers (EM1–EM3), and the Hadronic layers (HAD). As expected for electromagnetic showers initiated by electrons, the largest energy concentrations occur in the EM2 region, followed by smaller but non-negligible contributions in the PS and EM1 regions. Minimal energy is deposited in the EM3 and hadronic layers, consistent with the containment of the electromagnetic shower within the electromagnetic calorimeter.

This visualization highlights the characteristic energy deposition pattern across calorimeter layers, and illustrates how the *Ringer* algorithm captures radial information effectively in a compact 100-D feature space. These transformed features serve as the foundation for the subsequent analysis, where we evaluate various methods of feature selection and extraction to identify the most informative representations for energy calibration.

A. Feature Selection and Extraction Evaluation

Feature selection and extraction methods are analyzed through heatmaps shown in Fig. 8. These maps depict the correlation, mutual information, PCA, and ICA relationships with E_{true}. High-correlation regions depicted in dark blue are guiding the selection of attributes for ML methods.

Table III summarizes the sets of features selected or extracted by each dimensionality reduction method evaluated in this study. The first row (`Rings[1:100]`) represents the original input space, composed of all 100 normalized ring energies generated by the Lorenzetti Showers framework.

The next rows present the features retained by the correlation-based (CORR) and mutual information-based (MUTI) selection strategies under different thresholds. For instance, the configuration `CORR-50` includes all rings whose absolute Pearson correlation with the true energy exceeds 0.50,

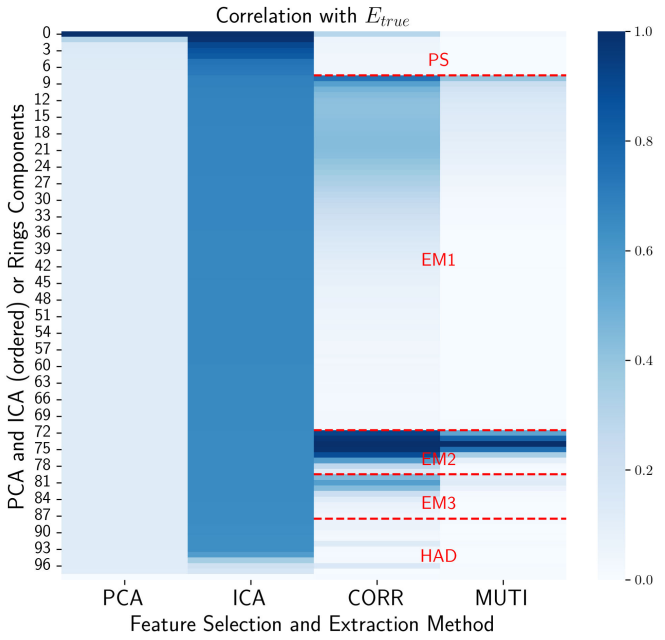


Fig. 8. Heatmap displaying the correlation between each of the 100 components and the true energy E_{true}, across four different feature selection and extraction methods: PCA, ICA, correlation (CORR), and mutual information (MUTI). Each column corresponds to one method, and each row represents a component index from 1 to 100. Darker blue shades indicate stronger correlation with E_{true}, while lighter shades (approaching white) indicate weaker correlation. The vertical red dashed lines mark the approximate boundaries of calorimeter regions: PS, EM1–EM3 (electromagnetic layers), and HAD (hadronic layers). These boundaries are relevant only for CORR and MUTI methods, which operate directly on the original ring energies; for PCA and ICA, the components are entirely new linear combinations of the original rings and therefore do not map directly to specific calorimeter regions. Variance analysis shows that more than 99% of the total variance is captured by the first ten components for both PCA and ICA.

TABLE III
SELECTED/EXTRACTED FEATURES AND NUMBER OF FEATURES (F) BY CORRELATION, MUTUAL INFORMATION, PCA, AND ICA APPROACHES

Selection/Extraction	Features	F
Rings[1:100]	Rings[1:100]	100
CORR-50	Rings[8, 9, 72, 73, 74, 75, 76, 77, 81]	9
CORR-70	Rings[8, 72, 73, 74, 75, 76]	6
CORR-80	Rings[72, 73, 74, 75, 76]	5
CORR-90	Rings[72, 73, 74, 75]	5
MUTI-50	Rings[72, 73, 74, 75]	4
MUTI-70	Rings[73, 74, 75]	3
MUTI-80	Rings[73, 74]	2
MUTI-90	Rings[74]	1
PCA-100	PCA[1:100]	100
PCA-10	PCA[1:10]	10
PCA-5	PCA[1:5]	5
PCA-2	PCA[1:2]	2
ICA-100	ICA[1:100]	100
ICA-10	ICA[1:10]	10
ICA-5	ICA[1:5]	5
ICA-2	ICA[1:2]	2

yielding a subset of nine informative features. As the threshold increases (e.g., CORR-90), the number of selected rings decreases, concentrating only on the most strongly correlated components (e.g., Rings[72, 73, 74, 75]).

Similarly, the MUTI-based method selects features according to the mutual information with the true energy. As expected, higher thresholds lead to fewer retained features, with MUTI-90 selecting only a single highly informative ring (Ring 74).

Feature selection using a 90% correlation threshold (CORR-90) and a 50% mutual information threshold (MUTI-50) identified the same attributes: Rings[72, 73, 74, 75]. This results in nearly identical statistics and graphs for these scenarios.

It is noteworthy that the strong correlation and mutual information observed for ring variables 72–78 can be attributed to their spatial arrangement within the calorimeter layers. These rings are positioned close to each other, leading to similar responses to incident particles due to shared energy deposition characteristics. This spatial relationship increases the likelihood of correlated measurements, as they capture similar energy distributions from particle showers.

For the PCA and ICA methods, which perform feature extraction rather than selection, the table lists the top $m \in \{100, 10, 5, 2\}$ components ranked by their absolute correlation with the target variable. For example, PCA-5 retains the first five principal components (PCA[1:5]), capturing the directions of highest variance aligned with the energy prediction task.

Overall, the table highlights how different dimensionality reduction methods reduce the feature space from 100 original inputs to as few as 1–10 components, enabling significant gains in inference speed and resource efficiency. This reduction is achieved without compromising predictive accuracy, as demonstrated in the subsequent performance evaluations.

The following sections present a detailed evaluation of the calibration performance for each configuration, using the error-based metrics established in the methodology. These analyses are crucial to assess how well the selected or extracted features support the estimation of the true energy in HEP environments.

B. Energy Error Evaluation

This section evaluates the performance of the feature selection and extraction methods (CORR, MUTI, PCA, and ICA) in combination with the ML regression models (MLR, PREG, DTREE, and BDT). The analysis is based on the calibration error ($\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$) and aims to compare the predictive accuracy achieved by each configuration.

To this end, we employ visualizations such as histograms and boxplots, along with statistical metrics including the IQR and RMSE, to characterize both the bias and variability of the calibration results. The corresponding results are presented in Fig. 9(a)–(d).

To better interpret the boxplots used in this evaluation, it is important to clarify their structure. The box width represents the IQR, while the black dot indicates the RMSE. The central line inside each box denotes the median (50th percentile) of the data, and the lower and upper edges correspond to the first quartile (25th percentile) and third quartile (75th percentile), respectively, defining the IQR. The whiskers extend from the box to the most extreme data points within 1.5 times the IQR from the quartiles. This visualization effectively captures the central tendency, dispersion, and potential outliers in the calibration error distributions.

1) Correlation (CORR): Fig. 9(a) presents the calibration error analysis for correlation-based feature selection

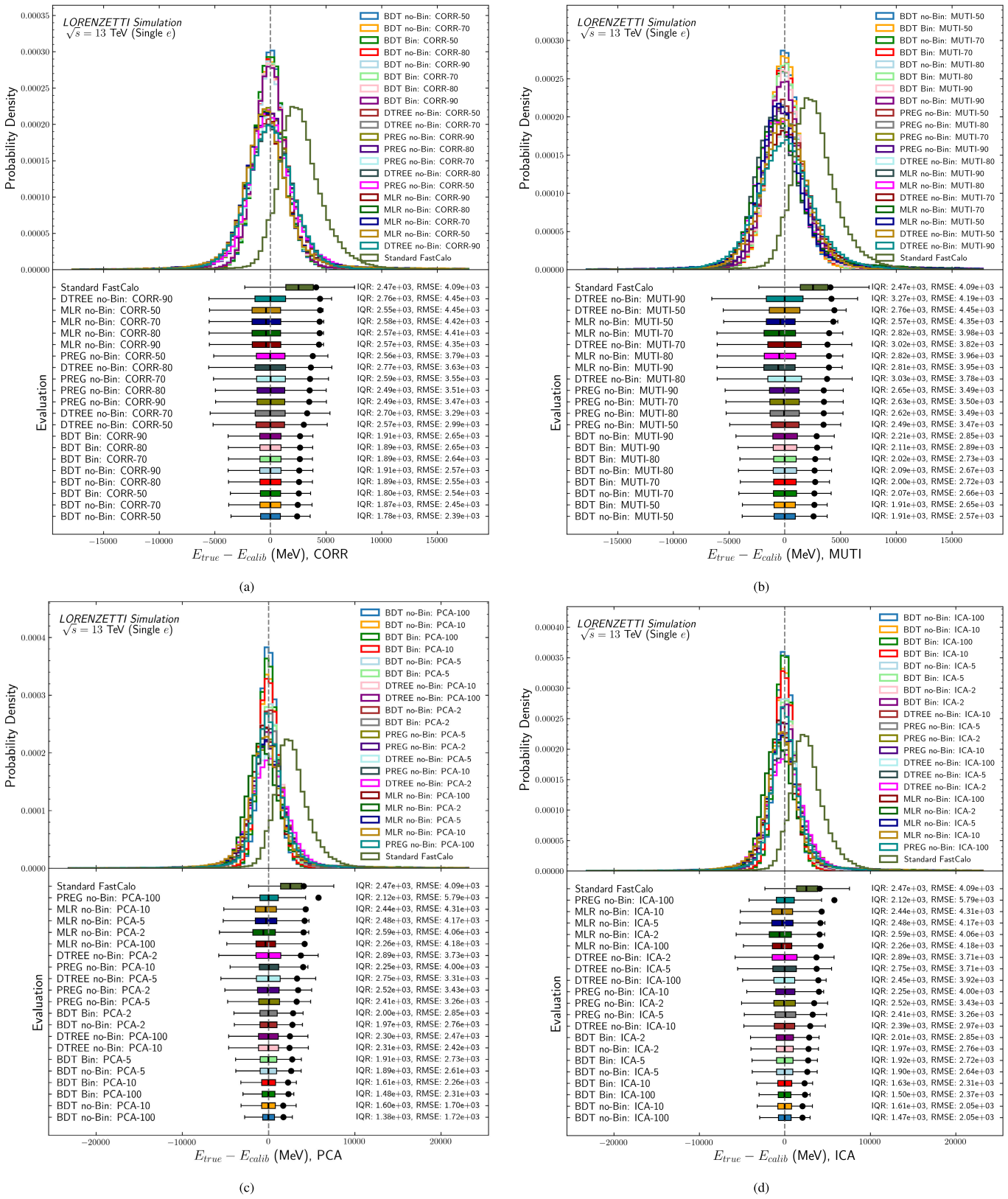


Fig. 9. Histograms of the $E_{\text{true}} - E_{\text{calib}}$ error (top) and corresponding boxplots (bottom) comparing the IQR and RMSE (indicated by black points) across different ML models. Each subplot corresponds to a distinct feature selection or extraction strategy. (a) Correlation-based selection, (b) mutual information-based selection, (c) PCA-based extraction, and (d) ICA-based extraction.

methods, ordered by the global d_0 metric [as defined in (22)]. The Standard FastCalo baseline exhibits the highest mean and median errors, along with a substantial deviation from zero, resulting in the largest RMSE and variance among all evaluated configurations. In contrast, the BDT models without binning, specifically those using correlation thresholds of 50%

TABLE IV

PERFORMANCE SUMMARY OF THE EVALUATED CALIBRATION METHODS, EXPRESSED IN TERMS OF IQR, RMSE, AND COMPOSITE METRIC d_0. ALL VALUES WERE COMPUTED USING THE INDEPENDENT TEST SET (30% OF THE DATASET, HELD OUT FROM TRAINING), ENSURING AN UNBIASED ESTIMATE OF PREDICTIVE PERFORMANCE. THE STANDARD FASTCALO (STD) SERVES AS THE BASELINE FOR ALL RELATIVE PERCENTAGES SHOWN

Method-Selection/Extraction	IQR (GeV)	RMSE (GeV)	$d0$ (GeV)	IQR (% Std)	RMSE (% Std)	$d0$ (% Std)
Standard FastCalo (Std)	2.47	4.09	4.78	0.00%	0.00%	0.00%
BDT no-Bin: PCA-ALL	1.38	1.72	2.21	44.13%	57.95%	53.85%
BDT no-Bin: PCA-10	1.60	1.70	2.33	35.22%	58.44%	51.14%
BDT no-Bin: ICA-ALL	1.47	2.05	2.52	40.49%	49.88%	47.20%
BDT no-Bin: ICA-10	1.61	2.05	2.61	34.82%	49.88%	45.44%
BDT no-Bin: CORR-50	1.78	2.39	2.98	27.94%	41.56%	37.63%
BDT no-Bin: CORR-70	1.87	2.45	3.08	24.29%	40.10%	35.49%
BDT no-Bin: MUTI-50	1.91	2.57	3.20	22.67%	37.16%	32.98%
BDT Bin: MUTI-50	1.91	2.65	3.27	22.67%	35.21%	31.63%

(CORR-50) and 70% (CORR-70), achieved the lowest IQR and RMSE values. These configurations utilize only nine and six features, as reported in Table III, highlighting the effectiveness of dimensionality reduction through correlation-based selection. Among all evaluated methods, `BDT no-Bin: CORR-50` delivered the best overall performance, achieving a 27.94% reduction in IQR and a 41.56% reduction in RMSE relative to the Standard FastCalo. This corresponds to a total improvement of 37.66% in the d_0 metric, as summarized in Table IV, demonstrating both improved calibration accuracy and efficiency.

2) Mutual Information (Muti): Fig. 9(b) presents the calibration error analysis for feature selection methods based on mutual information, with configurations ranked according to the global d_0 metric. As observed in the correlation-based evaluation, the Standard FastCalo energy also shows the highest mean and median calibration errors, along with the largest RMSE, reinforcing its role as a conservative baseline. Among the evaluated configurations, BDT models using a mutual information threshold of 50% (MUTI-50), both with and without binning, demonstrated the best overall performance. In particular, the `BDT no-Bin: MUTI-50` configuration, which employs only four features (see Table III), achieved an IQR reduction of 22.67% and an RMSE reduction of 37.16% compared to the Standard FastCalo baseline. This corresponds to a total improvement of 32.98% in the d_0 metric, as summarized in Table IV. While these results highlight the effectiveness of mutual information for dimensionality reduction, it is worth noting that the best MUTI-based configuration still falls slightly short of the top-performing correlation-based model in terms of overall calibration accuracy.

3) Principal Component Analysis: Fig. 9(c) displays the calibration error analysis for PCA-based feature extraction methods. As in previous evaluations, the Standard FastCalo configuration presents the highest RMSE and serves as a conservative reference point for comparison. Among the tested models, BDT regressors without binning, using either all 100 principal components (PCA-100) or the top 10 components (PCA-10), achieved the lowest calibration errors. Specifically, the `BDT no-Bin: PCA-100` configuration reduced the IQR by 44.13% and the RMSE by 57.95%, while `BDT no-Bin: PCA-10` achieved a 35.22% reduction in IQR and an even greater 58.44% reduction in RMSE, both relative to the Standard FastCalo, as summarized in Table IV. These improvements correspond to global d_0 metric reductions of 53.85% and 51.14% for `BDT no-Bin: PCA-100` and `BDT no-Bin: PCA-10`, respectively. Notably, both configurations outperform all previous methods, including the top correlation-based and mutual information-based models, highlighting PCA's effectiveness in capturing the most relevant variance directions for energy calibration tasks.

4) Independent Component Analysis: Fig. 9(d) presents the calibration error analysis for ICA-based feature extraction methods. As observed in previous evaluations, the Standard FastCalo configuration yields the highest calibration errors and variance, reinforcing its role as a baseline. The best-performing ICA configurations were obtained using BDT regressors without binning, applied to either all 100 components (ICA-100) or the top 10 components (ICA-10). The `BDT no-Bin: ICA-100` model achieved a 40.49% reduction in IQR and a 49.88% reduction in RMSE relative to the Standard FastCalo, outperforming all correlation- and mutual information-based methods. However, it still fell short of the top-performing PCA configurations. Similarly, the `BDT no-Bin: ICA-10` setup reduced the IQR by 34.82% and the RMSE by 49.88%, offering a substantial improvement in predictive accuracy while using significantly fewer features. Although these results confirm the effectiveness of ICA in capturing statistically independent sources of information, they do not surpass the performance achieved by PCA-based models.

5) General Evaluation: Table IV summarizes the performance of the best-performing methods, ranked in ascending order by the composite metric d_0, which integrates both RMSE and IQR to provide a comprehensive measure of calibration accuracy. The Standard FastCalo configuration, listed at the top, serves as the baseline reference for all comparisons.

All alternative configurations based on BDT regressors exhibit substantial improvements in both error dispersion (IQR) and average calibration error (RMSE). Notably, the `BDT no-Bin: PCA-100` method achieves the best overall result, with a 53.85% reduction in the d_0 metric, alongside a 44.13% decrease in IQR and a 57.95% reduction in RMSE compared to the standard method.

A similarly strong performance is observed for `BDT no-Bin: PCA-10`, which achieves the lowest RMSE (58.44% reduction) among all methods while requiring significantly fewer input features, making it a particularly attractive option for real-time deployment scenarios.

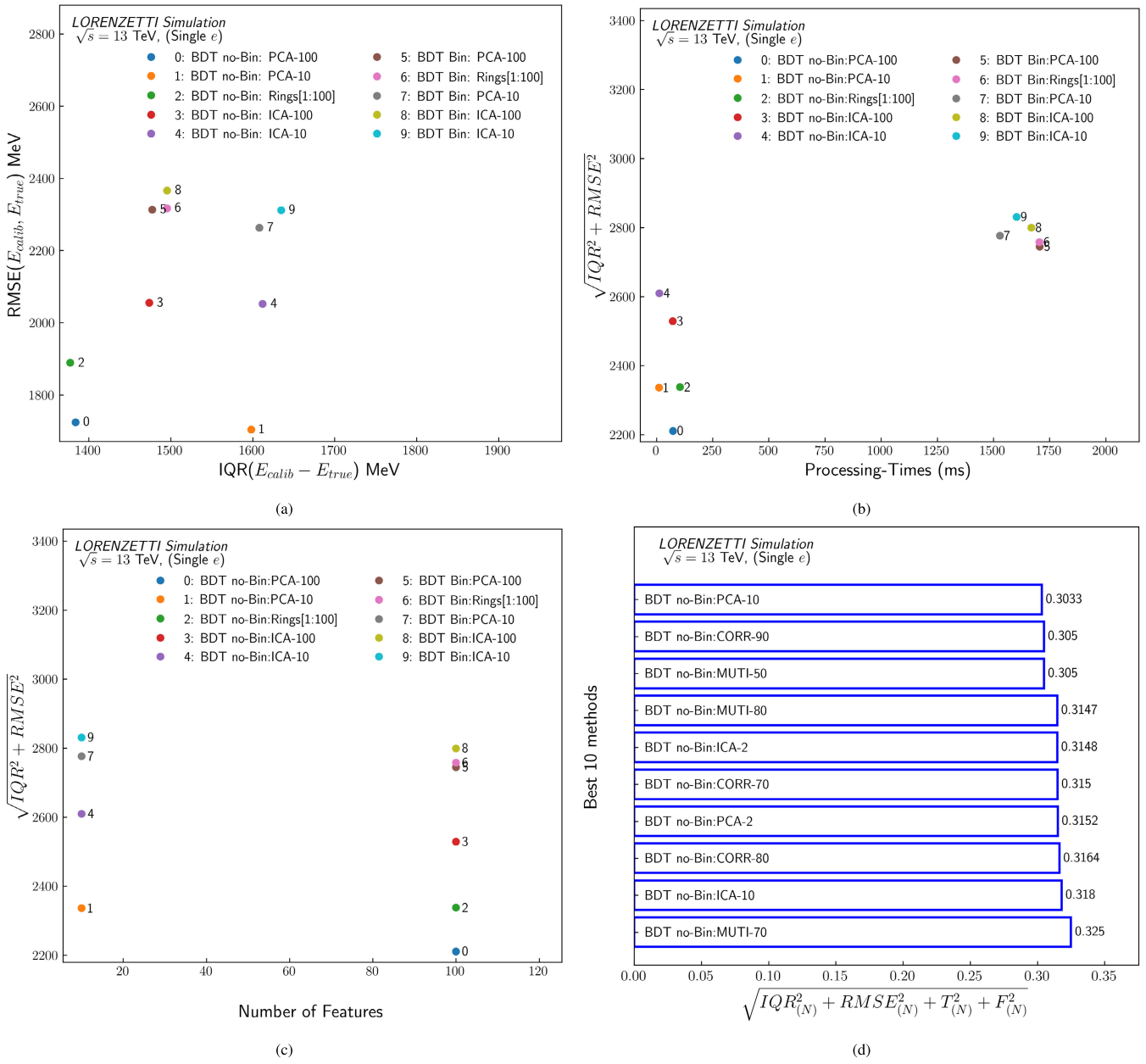


Fig. 10. Comparative analysis of the top ten combinations of feature selection/extraction techniques and ML models. (a) Tradeoff between RMSE and IQR, highlighting the overall error distribution. (b) Discriminative power metric d_0 against processing time, illustrating the balance between accuracy and computational cost. (c) d_0 versus the number of input features, emphasizing model efficiency relative to dimensionality. (d) Overall ranking of all methods based on their d_0 performance.

Other configurations, such as `BDT no-Bin: ICA-100`, `BDT no-Bin: CORR-50`, and `BDT no-Bin: MUTI-50`, also demonstrate notable performance gains. All BDT-based approaches evaluated in this table achieved reductions of more than 30% in the d_0 metric relative to the Standard FastCalo, confirming the effectiveness of combining dimensionality reduction techniques with ML models for energy calibration.

These results highlight the potential of data-driven calibration strategies to significantly enhance both accuracy and robustness in high-throughput, trigger-level systems.

Additionally, Fig. 10(a) provides a 2-D visualization of the IQR and RMSE values for the ten best-performing configurations. The three methods that achieved the lowest IQR—indicating the smallest dispersion in calibration errors—were: 1) `BDT no-Bin: PCA-ALL`; 2) `BDT no-Bin: Rings[1:100]`; and 3) `BDT no-Bin: ICA-ALL`. In terms of RMSE, which reflects the average magnitude of the calibration error, the top-performing configurations were: 1) `BDT no-Bin: PCA-10`; 2) `BDT no-Bin: PCA-ALL`; and 3) `BDT no-Bin: Rings[1:100]`. It is worth noting that the configuration using the full set of normalized energy rings (`Rings[1:100]`) was included in the comparison as a reference for evaluating the impact of dimensionality reduction techniques on ML regressors. Its recurrent appearance among the best results underscores its strong baseline performance despite the absence of feature selection or transformation. Overall, these findings reinforce the effectiveness of PCA-based dimensionality reduction—particularly when retaining

ten or more components—and highlight the competitive accuracy and stability offered by the uncompressed ring profile.

The systematic error observed in the standard FastCalo approach, as shown in Fig. 9(a)–(d), stems from multiple factors, including electronic noise, pile-up effects from multiple collisions, and intrinsic limitations of the calorimeter response. As evidenced in Fig. 4(b), nonlinearities are present in the simulation itself, as seen in the scatter plot of E_{fast} versus E_{true}, particularly from middle to higher energy ranges. To mitigate these issues, we introduced a calibration factor α_{calib} that adjusts the reconstructed transverse energy E_{fast} to better approximate the true transverse energy E_{true}. As stated before, this calibration is performed by minimizing the mse between E_{calib} and E_{true}, thereby improving the energy estimates and correcting for systematic biases inherent in the Standard FastCalo energy.

Furthermore, the step-like behavior observed in E_{fast} above 250GeV in Fig. 4(b) can be attributed to the combined effect of increased energy estimation accuracy with rising energy, an expected characteristic of calorimetric response [56], and the application of conversion coefficients optimized for lower energy particles. Despite this deviation in the high-energy regime, the ML calibration methods maintained strong performance, achieving lower RMSE and IQR values compared to the standard estimations. These results indicate that the ML-based approaches are robust even when exposed to systematic variations across the energy spectrum.

Moreover, the decision to limit the number of retained components to 2, 5, or 10 is motivated not only by the explained variance analysis but also by practical constraints associated with real-time experimental applications. In these contexts, energy calibration must be performed with low latency and minimal computational overhead, which necessitates the use of less complex models. From a statistical standpoint, PCA results show that the first two components alone account for approximately 99.71% of the total variance in the dataset, indicating that most of the relevant information is concentrated in a very low-dimensional subspace. Similarly, ICA displays analogous behavior, with the majority of the independent structure captured by the first few components. As such, the inclusion of additional components would result in marginal informational gain at the expense of increased computational cost.

Therefore, the selected dimensionality limits represent a deliberate trade-off between model simplicity and performance. This strategy effectively reduces the risk of overfitting while maintaining high calibration accuracy, an essential requirement for high-throughput environments where model generalization is critical. As shown in Table IV, increasing the number of components from 10 to 100 leads to only a marginal improvement of approximately 2% in the d_0 metric for both PCA- and ICA-based methods. This further supports the conclusion that retaining ten or fewer components is sufficient for near-optimal calibration performance under the targeted operational constraints.

To more thoroughly evaluate the computational cost associated with each method, we now analyze how the number of features and inference time influence the overall ranking of the calibration strategies.

C. Number of Features and Processing Time Evaluation

In scenarios where processing speed and memory efficiency are critical—such as real-time data acquisition systems—the number of input features (F) and the inference time per event (T) become decisive factors in selecting the most appropriate calibration method. Notably, the best-performing configurations rely on all 100 attributes generated by PCA transformations. This observation motivates a detailed comparison of the feature count and processing time to identify methods that achieve the best trade-off between predictive performance and computational efficiency.

Fig. 10(b) and (c) depicts the relationship between the d_0 metric and processing time (T) and number of features (F) across all evaluated methods.

Fig. 10(b) illustrates the relationship between processing time and the composite performance metric d_0. Methods labeled 0–4 stand out by achieving both low inference times and the lowest d_0 values, reflecting superior calibration accuracy with high computational efficiency. In contrast, several methods (5)–(9) exhibit very short processing times but significantly higher d_0 values, indicating a trade-off with reduced predictive performance. Among the top-performing configurations, `BDT no-Bin: PCA-10` and `BDT no-Bin: ICA-10` (methods 1 and 4, respectively) achieved the shortest inference times, reinforcing their suitability for latency-constrained environments.

Fig. 10(c) reveals that methods 1, 4, 7, and 9 deliver calibration performance comparable to the best-ranked models in terms of the composite d_0 metric while employing fewer than ten input features. Among them, method 1 (`BDT no-Bin: PCA-10`) stands out by achieving high predictive accuracy with a minimal feature set, reinforcing its practicality for real-time and resource-constrained environments.

Fig. 10(d) presents a revised ranking based on the global metric D_0 [defined in (23)], which integrates accuracy metrics (IQR and RMSE), inference time (T), and number of features (F). As illustrated, the top-performing methods are those that simultaneously achieve low calibration error, rapid inference, and compact feature representations. These findings highlight the effectiveness of feature-efficient models in delivering high-precision energy calibration with minimal computational overhead, reinforcing their applicability in latency-sensitive trigger-level systems. It is important to emphasize that several methods appearing in this new ranking were not present among the top performers in Table IV, which was based solely on the d_0 metric that considers only IQR and RMSE. Their emergence in the D_0-based ranking reflects the equal weighting now given to processing time and feature count. Therefore, Fig. 10(d) represents not a continuation, but a complementary perspective, offering a more holistic evaluation of the trade-offs between predictive performance and computational efficiency.

The most favorable trade-offs between accuracy (IQR and RMSE), processing time (T), and model complexity (number of features F), as captured by the composite $D0$ metric, are obtained by the top-ranked configurations presented in Fig. 10(d) and summarized in Table V.

1) `BDT no-Bin: PCA-10` delivers the best overall performance, achieving the lowest $D0 = 0.3032$ with only ten features and moderate processing time (9.35 ms).
2) `BDT no-Bin: CORR-90` offers a very competitive trade-off, using only four features and achieving fast inference time (3.29 ms), with $D0 = 0.3049$.
3) `BDT no-Bin: MUTI-50` matches `BDT no-Bin: CORR-90` in both number of features (4) and overall metric ($D0 = 0.3049$), with a slightly higher processing time (4.35 ms).
4) `BDT no-Bin: MUTI-80` and `BDT no-Bin: ICA-2` also stand out by using only two features and maintaining low latency (2.22 and 1.91 ms, respectively), with $D0 = 0.3147$.

These results are significant in terms of both data storage and processing efficiency. Using a substantially reduced number of features not only improved performance compared to the Standard FastCalo measurement but also minimized the computational and storage requirements of the calibration process. Fewer input features result in smaller data representations, which directly reduce the amount of memory and disk space needed for storing and handling event-level information. In addition, the reduced dimensionality leads to faster model inference and shorter processing times for ML-based calibration methods.

Among the tested configurations, the four top-ranked methods listed above exhibited prediction times below 10 ms when evaluated on a standard laptop with limited hardware resources. This demonstrates their computational feasibility even in nonspecialized environments. Therefore, when deployed on high-performance computing systems, such as those available in real-time trigger infrastructures of large-scale HEP experiments (e.g., ATLAS at the LHC), these methods are expected to achieve even lower latencies, potentially meeting the stringent millisecond requirements imposed by such environments [17].

Importantly, these benefits were achieved using only a compact set of features, allowing irrelevant attributes to be discarded early in the pipeline. This strategy not only enhances computational performance but also contributes to a leaner and more scalable data-processing chain.

The subsequent sections will present results related to the calibration factor in the used binning.

D. Calibration Factor per Bin Evaluation

The calibration factors (α_{calib}) produced by the top four methods, ranked according to the D_0 metric, were evaluated and compared against the ideal reference values $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$, derived from the Standard FastCalo measurements. This evaluation employs the IQR, MAE, and RMSE to quantify the deviation of α_{calib} from α_{true} across bins defined in terms of pseudorapidity (η) and transverse energy (E_t).

Beginning with the IQR analysis, Fig. 11(a) and (b) displays the spread of calibration factors within the η and E_t bins, respectively.

1) *Increasing η:* Fig. 11(a) shows that the IQR of calibration factors increases with $|\eta|$, reflecting greater uncertainty as $|\eta|$ moves away from zero, where precision is highest.
2) *Increasing Energy:* Fig. 11(b) indicates that IQR decreases with higher energy due to a better signal-to-noise ratio, which reduces calibration uncertainty.
3) *Method Overlap:* The methods `BDT no-Bin: CORR-90` and `BDT no-Bin: MUTI-50` produce overlapping results, as they use the same set of rings.

For MAE and RMSE in estimating α_{calib}, Fig. 12(a) and (b) displays the results for various methods and bins.

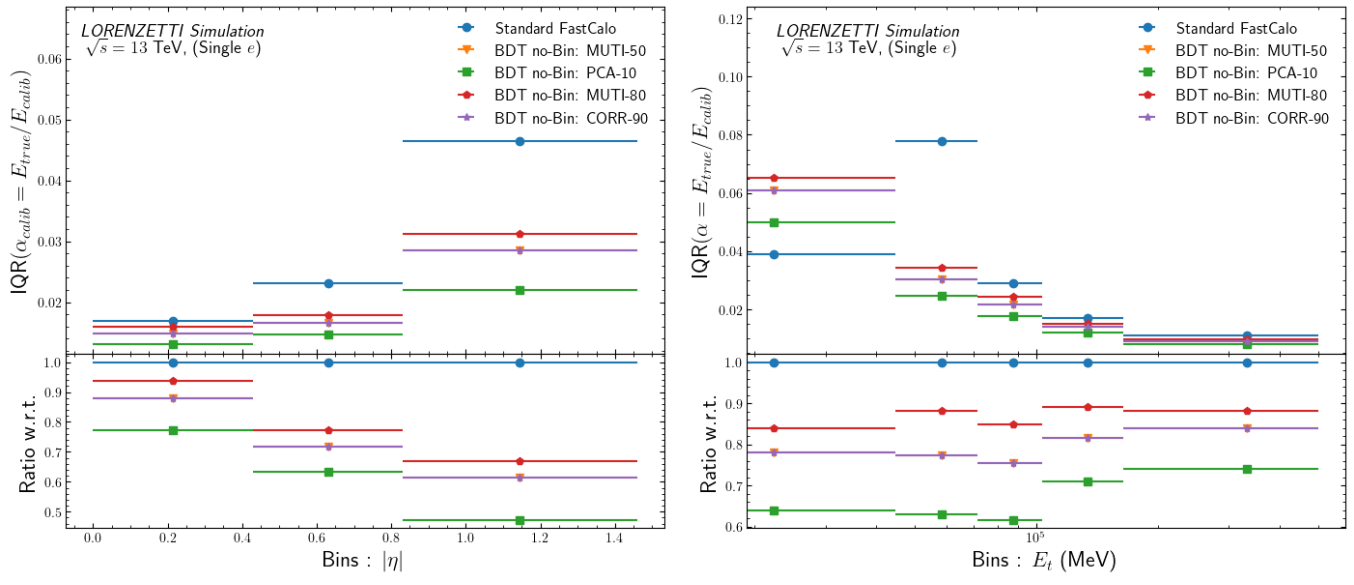
1) *Increasing η:* Fig. 12(a) confirms that MAE and RMSE increase with $|\eta|$, consistent with increased uncertainty for higher $|\eta|$ values.
2) *Increasing Energy:* Fig. 12(b) shows a reduction in MAE and RMSE with increasing energy, aligned with the improved signal-to-noise ratio.
3) As before, the results for `BDT no-Bin: CORR-90` and `BDT no-Bin: MUTI-50` overlap due to the same rings used for calibration estimation.

Finally, Fig. 13(a) and (b) presents the boxplots of α_{calib} values for comparison with the Standard FastCalo:

1) *Increased Dispersion With η:* Fig. 13(a) confirms the increased dispersion (as measured by IQR) with rising $|\eta|$. The implemented methods, however, align with the median of the Standard FastCalo measurement.
2) *Decreased Dispersion With Energy:* Fig. 13(b) confirms reduced dispersion (IQR) with increased energy, with implemented methods aligning in median with the Standard FastCalo measurement.
3) *Method Equality:* Results for `BDT no-Bin: CORR-90` and `BDT no-Bin: MUTI-50` are identical due to using the same rings for calibration estimation.

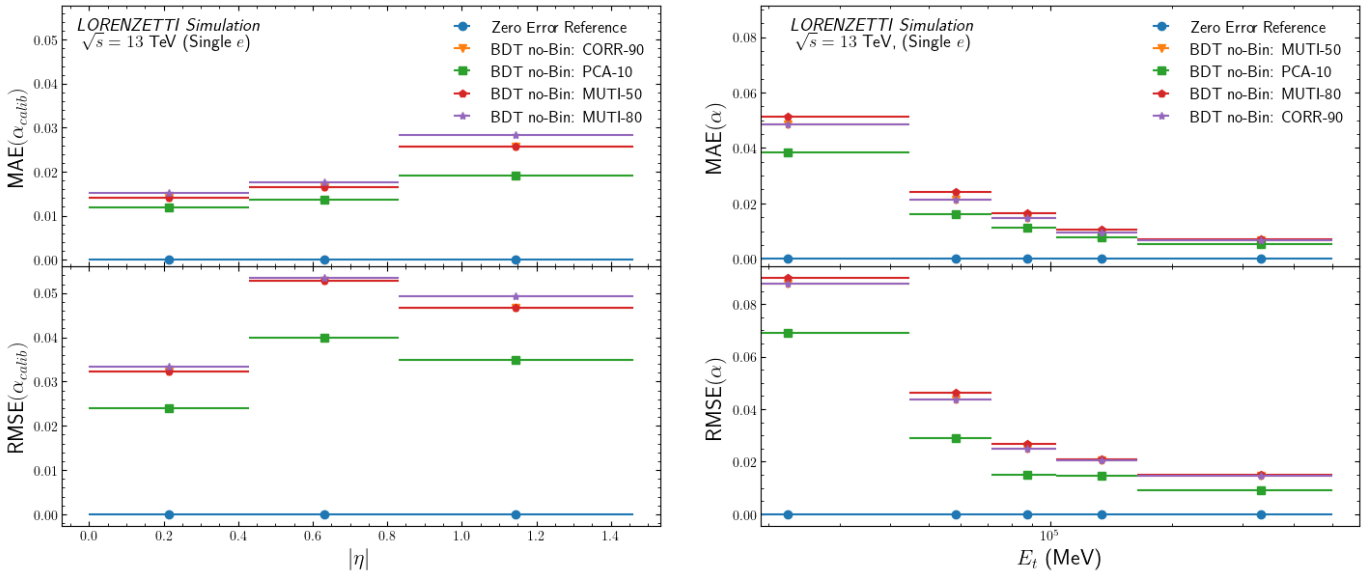
The methods MLR, PREG, and DTREE were evaluated without binning due to the underlying assumptions of each model and the potential issues introduced by binning. Although PREG (polynomial regression) can capture nonlinearities, both PREG and MLR assume functional continuity in the input domain. Binning the data by η and E_T may introduce discontinuities at bin boundaries, which could lead to instability or loss of generalization capacity—especially for models sensitive to smooth transitions. While this behavior was not explicitly tested for these methods, binning was evaluated using the best-performing method (BDT) as a controlled case.

The methods MLR, PREG, and DTREE were initially evaluated without binning to maintain a consistent statistical basis across all models. While methods such as PREG (polynomial regression) and MLR assume functional continuity and may, in principle, be affected by discontinuities introduced by binning in η and E_T, no intrinsic limitation prevented the use of binning for any of the models. Instead, all methods were first tested on the full dataset without segmentation, and the best-performing one (BDT) was subsequently reevaluated with binning as a controlled comparison. The binned BDT did not surpass the performance of its non-binned version, likely due to reduced statistics in each segment. Therefore, binning was applied only to the BDT model for comparative analysis,



(a) (b)

Fig. 11. Comparison of the calibration factor IQR(α) across different (a) $|\eta|$ and (b) transverse energy (E_T) bins. The y-axis represents the ratio $E_{\text{true}}/E_{\text{calib}}$, where E_{calib} includes both the calibrated energy estimates and the Standard FastCalo (E_{fast}). The upper section of each plot displays absolute values, while the lower section presents values *with respect to* (i.e., relative to) the Standard FastCalo performance. In this representation, the Standard FastCalo is fixed at 1.0 for all bins, and the curves for other calibration methods show deviations around this baseline, indicating how far each method's IQR(α) differs from the standard reference across $|\eta|$ and E_T variations.

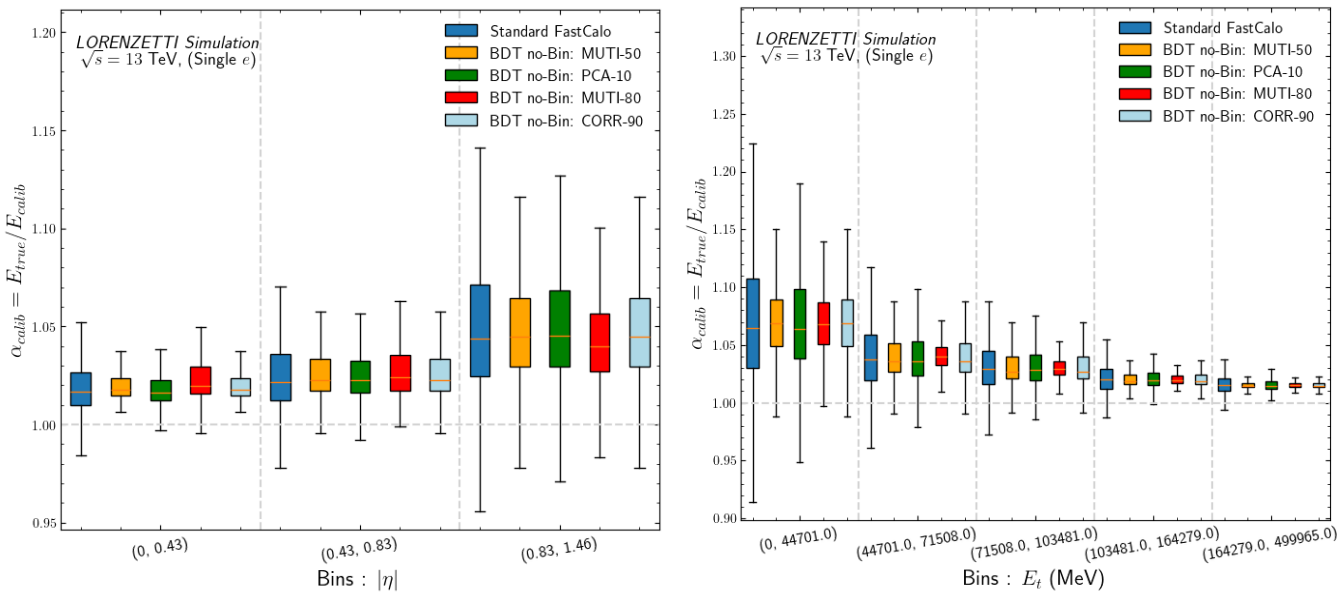


(a) (b)

Fig. 12. Comparison of the MAE and RMSE of α_{calib} estimates across different (a) $|\eta|$ and (b) transverse energy (E_T) bins. The upper y-axis represents the MAE(α), while the lower y-axis represents the RMSE(α). The calibration error for the Standard FastCalo here is assumed to be zero, as it was used as the target during the ML training process, and it corresponds directly to the ratio $E_{\text{true}}/E_{\text{fast}}$.

while the other methods were retained in their original non-binned configurations to preserve statistical robustness and model comparability. Due to this limitation, the binned BDT was not ranked among the top-performing methods. A more complete evaluation of segmentation-based approaches is left for future work, where a larger simulated dataset will allow for a more robust and statistically reliable analysis across segmented regions.

Beyond computational gains, these improvements translate into practical advantages for real-time event selection. Fig. 14 illustrates a trigger turn-on curve, where $\Delta\lambda$ represents the average energy threshold, centered on the transverse energy bin targeted for event selection. It defines the range of events that will be accepted or rejected based on their energy. In the figure, it can be observed that the *BDT no-Bin: PCA-10* configuration achieves a $\Delta\lambda$ of 2053.73 MeV (blue curve) for 90% event acceptance, whereas the Standard FastCalo exhibits a $\Delta\lambda$ of 5509.94 MeV (orange curve) for the same acceptance rate. This result demonstrates that, with a lower threshold in the event selection menu, the `BDT no-Bin: PCA-10`



(a) (b)

Fig. 13. Calibration factor α_{calib} boxplot in (a) η and (b) E_t bins. The width of the boxes represents the IQR. The gray dashed lines delimit the regions defined by the binning in η and E_t.

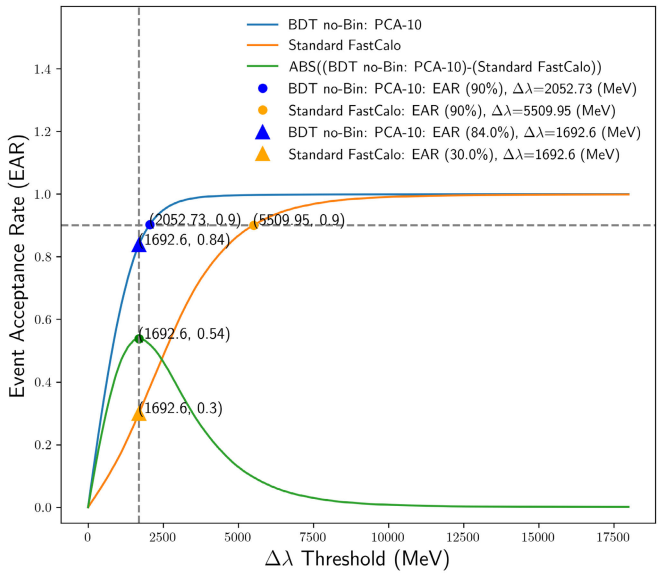


Fig. 14. Trigger turn-on curve comparing event acceptance performance between BDT no-Binning: PCA-10 (blue) and Standard FastCalo (orange). The BDT method achieves 90% acceptance with a lower energy threshold ($\Delta\lambda$ = 2053.73 MeV), compared to 5509.94 MeV for FastCalo. The green curve shows the absolute acceptance gain across energy bins, highlighting the improved efficiency of the BDT approach in selecting events at lower energies.

is able to maintain a high acceptance rate. Similarly, the green curve shows the absolute difference in event acceptance between the BDT no-Binning: PCA-10 and the Standard FastCalo. The peak of this curve represents the maximum difference between the two methods. At this peak, the BDT no-Binning: PCA-10 maintains an 85% acceptance rate with a $\Delta\lambda$ of 1692.6 MeV, while the Standard FastCalo only achieves a 30% acceptance rate. This confirms the capability of the `BDT no-Bin: PCA-10` to accept a larger fraction of events with a lower energy threshold.

These results provide insight into the calibration factor performance across different bins and event acceptance rates, highlighting the effectiveness of the top-ranked methods.

TABLE V

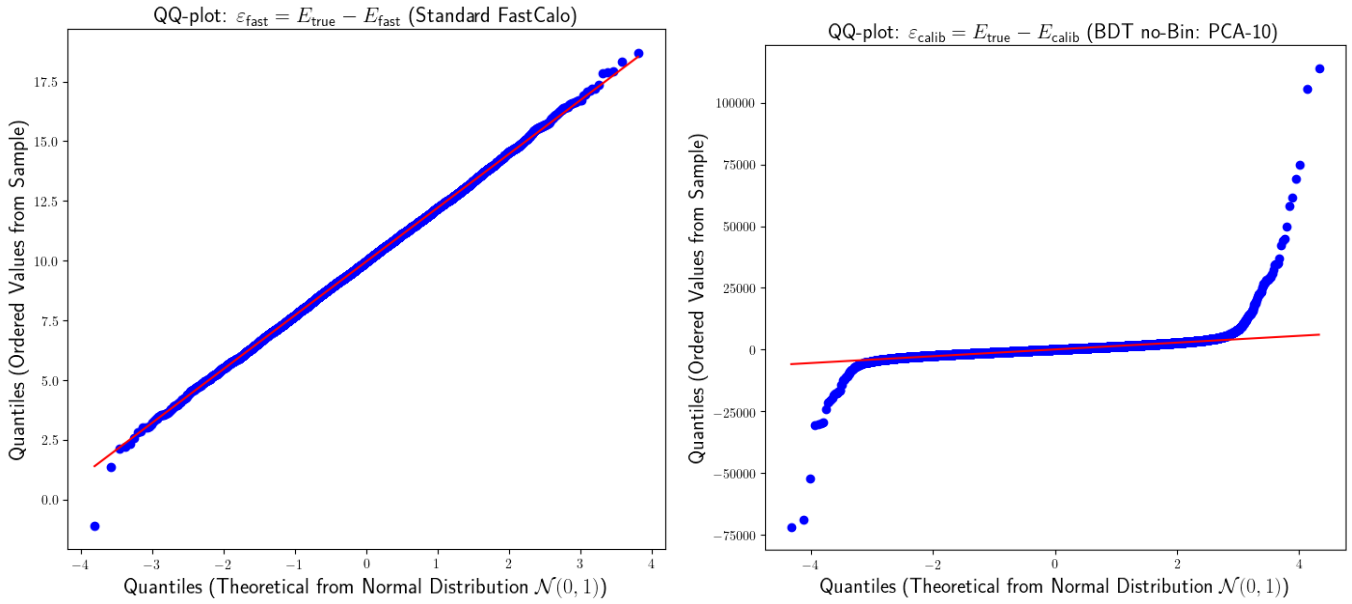
TOP FIVE METHODS ACCORDING TO THE COMPOSITE D_0 METRIC, WHICH JOINTLY ACCOUNTS FOR ERROR METRICS (d_0 COMPUTED FROM IQR AND RMSE), INFERENCE TIME PER EVENT (T), AND NUMBER OF INPUT FEATURES (F). ALL PERFORMANCE VALUES ARE COMPUTED USING THE INDEPENDENT 30% TEST SET, WITH NO ADDITIONAL CROSS-VALIDATION PROCEDURES APPLIED. THE RELATIVE IMPROVEMENT IN d_0 IS SHOWN WITH RESPECT TO THE STANDARD FASTCALO BASELINE

Method	$d0$ **(GeV)**	$d0$ **(% Std)**	T **(ms)**	F
Standard FasCalo (Std)	4.78	0.00%	-	-
BDT no-Bin: PCA-10	2.33	51.14%	9.35	10
BDT no-Bin: Corr-90	3.20	33.03%	3.29	4
BDT no-Bin: MutI-50	3.20	32.98%	4.35	4
BDT no-Bin: MutI-80	3.39	29.05%	2.22	2
BDT no-Bin: ICA-2	3.39	29.05%	1.91	2

E. Statistical Validation

The normality (Gaussianity) assessment for $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$ and $\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$, as well as for their paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$, using the D'Agostino–Pearson and Anderson–Darling tests, yielded p-values $< 10^{-16}$ in all configurations. This led to rejection of the null hypothesis of normality for every distribution examined. Representative Q–Q plots for the BDT no-Bin PCA-10 configuration (Fig. 15) show marked deviations from the $y = x$ line in both tails, consistent with heavy-tailed and asymmetric behavior, a pattern also observed for the other top-ranked methods.

Although the residual distributions are non-Gaussian, the data are strictly paired, allowing valid use of the paired tests



(a) (b)

Fig. 15. Representative Q–Q plots for (a) $\varepsilon_{\text{fast}}$ and (b) $\varepsilon_{\text{calib}}$ (BDT no-Bin: PCA-1) with respect to a standard normal distribution. Significant deviations from Gaussianity are visible in the distribution tails, indicating heavy-tailed and asymmetric behavior. This departure from normality motivates the adoption of complementary non-parametric or robust paired statistical methods, such as the Wilcoxon signed-rank test for central tendency and bootstrap-based inference for variance ratios, in addition to classical parametric approaches.

TABLE VI

STATISTICAL EVALUATION OF MEAN CALIBRATION ERROR USING PAIRED ONE-SIDED t, z, AND WILCOXON SIGNED-RANK TESTS. HYPOTHESES: $H_0 : \mu_d \geq 0$ VERSUS $H_1 : \mu_d < 0$, WHERE μ_d IS THE MEAN (OR MEDIAN FOR WILCOXON) OF THE PAIRED DIFFERENCES $d = \varepsilon_{\text{CALIB}} - \varepsilon_{\text{FAST}}$, WITH $\varepsilon_{\text{CALIB}} = E_{\text{TRUE}} - E_{\text{CALIB}}$ AND $\varepsilon_{\text{FAST}} = E_{\text{TRUE}} - E_{\text{FAST}}$. ALL CIS ARE ONE-SIDED AT 95%. UNITS FOR μ_d AND CIS ARE IN MEV

Method (z-test: primary)	z-statistic	p-value	95% one-sided CI for μ_d [MeV]	Decision
BDT no-Bin: PCA-10	-351.7106	$< 10^{-300}$	(-∞, -2731.0411]	Reject H0
BDT no-Bin: CORR-90	-540.8462	$< 10^{-300}$	(-∞, -2688.4327]	Reject H0
BDT no-Bin: MUTI-80	-623.1838	$< 10^{-300}$	(-∞, -2643.9968]	Reject H0
BDT no-Bin: MUTI-50	-540.8462	$< 10^{-300}$	(-∞, -2688.4327]	Reject H0
Method (t-test: reference)	t-statistic	p-value	95% one-sided CI for μ_d [MeV]	Decision
BDT no-Bin: PCA-10	-351.7106	$< 10^{-300}$	(-∞, -2731.0410]	Reject H0
BDT no-Bin: CORR-90	-540.8462	$< 10^{-300}$	(-∞, -2688.4327]	Reject H0
BDT no-Bin: MUTI-80	-623.1838	$< 10^{-300}$	(-∞, -2643.9968]	Reject H0
BDT no-Bin: MUTI-50	-540.8462	$< 10^{-300}$	(-∞, -2688.4327]	Reject H0
Method (Wilcoxon signed-rank test: robustness check)	W-statistic	p-value	95% one-sided CI for median(d) [MeV]	Decision
BDT no-Bin: PCA-10	72282864.0000	$< 10^{-300}$	(-∞, -2479.0331]	Reject H0
BDT no-Bin: CORR-90	46753096.0000	$< 10^{-300}$	(-∞, -2431.5419]	Reject H0
BDT no-Bin: MUTI-80	46212373.0000	$< 10^{-300}$	(-∞, -2497.2520]	Reject H0
BDT no-Bin: MUTI-50	46753096.0000	$< 10^{-300}$	(-∞, -2431.5419]	Reject H0

described in Section II-I. Non-parametric methods were also applied to confirm robustness against distributional departures, including the Wilcoxon signed-rank test for central tendency and the paired bootstrap BCa test for variance ratios.

1) Mean Error Comparison: Table VI summarizes the results of three complementary approaches applied to the paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$: the large-sample paired z-test (primary), the paired Student's t-test (reference), and the Wilcoxon signed-rank test (robustness check).

Across all configurations and test types, the null hypothesis $H_0 : \mu_d \geq 0$ was rejected at $\alpha = 0.05$ with overwhelming statistical significance. For both the z- and t-tests, the p-values were below the numerical precision of the machine, reported as $p < 10^{-300}$. The Wilcoxon signed-rank test, which does not require normality, yielded consistent rejections. In all cases, the one-sided 95% CIs for μ_d (or for the median of d in the Wilcoxon test) lay entirely below zero, which is the interval-based equivalent of rejecting H_0.

From a physical standpoint, negative values of μ_d indicate that calibration systematically reduces the residuals compared to the uncalibrated method. For `BDT no-bin: PCA-10`, $\mu_d = -2.73$ GeV (Table VI), meaning that, on average, the residual after calibration is at least 2.73 GeV closer to zero for the same event. Comparable magnitudes of improvement in μ_d are observed for `BDT no-bin: CORR-90`, `BDT no-bin: MUTI-80`, and `BDT no-bin: MUTI-50`, confirming that the effect is consistent across all ranked configurations.

2) Variance Ratio Analysis: Table VII presents the results of three complementary approaches for testing the variance ratio $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$: the Pitman–Morgan test (primary), the

TABLE VII

STATISTICAL EVALUATION OF THE ERROR VARIANCE RATIO USING PITMAN–MORGAN, ONE-SIDED F, AND PAIRED BOOTSTRAP TESTS. HYPOTHESES: $H_0 : \sigma^2_{\text{CALIB}}/\sigma^2_{\text{FAST}} \geq 1$ VERSUS $H_1 : \sigma^2_{\text{CALIB}}/\sigma^2_{\text{FAST}} < 1$, WITH σ^2_{CALIB} DENOTING THE VARIANCE OF $\varepsilon_{\text{CALIB}}$ AND σ^2_{FAST} THE VARIANCE OF $\varepsilon_{\text{FAST}}$. ALL CIS ARE ONE-SIDED AT 95% AND EXPRESSED IN DIMENSIONLESS UNITS (VARIANCE RATIOS). LOWER BOUNDS OF CIS WERE TRUNCATED AT 0 TO SATISFY THE NON-NEGATIVITY OF VARIANCE

Method (Pitman-Morgan: primary)	PM-statistic	p-value	95% one-sided CI for σ^2_{calib} / σ^2_{fast}	Decision
BDT no-Bin: PCA-10	-229.4583	$< 10^{-300}$	[0.0, 0.3472]	Reject H0
BDT no-Bin: CORR-90	-92.9690	$< 10^{-300}$	[0.0, 0.7642]	Reject H0
BDT no-Bin: MUTI-80	-82.6859	$< 10^{-300}$	[0.0, 0.8147]	Reject H0
BDT no-Bin: MUTI-50	-92.9690	$< 10^{-300}$	[0.0, 0.7642]	Reject H0
Method (F-test: reference)	**F-statistic**	**p-value**	**95% one-sided CI for σ^2_{calib} / σ^2_{fast}**	**Decision**
BDT no-Bin: PCA-10	0.3230	$< 10^{-300}$	[0.0, 0.3265]	Reject H0
BDT no-Bin: CORR-90	0.7349	$< 10^{-300}$	[0.0, 0.7430]	Reject H0
BDT no-Bin: MUTI-80	0.7914	$< 10^{-300}$	[0.0, 0.8001]	Reject H0
BDT no-Bin: MUTI-50	0.7349	$< 10^{-300}$	[0.0, 0.7430]	Reject H0
Method (Paired bootstrap BCa test: robustness check)	**(σ^2_{calib} / σ^2_{fast})-statistic**	**p-value**	**95% one-sided CI for σ^2_{calib} / σ^2_{fast}**	**Decision**
BDT no-Bin: PCA-10	0.3230	0.0002	[0.0, 0.3469]	Reject H0
BDT no-Bin: CORR-90	0.7349	0.0002	[0.0, 0.7649]	Reject H0
BDT no-Bin: MUTI-80	0.7914	0.0002	[0.0, 0.8149]	Reject H0
BDT no-Bin: MUTI-50	0.7349	0.0002	[0.0, 0.7649]	Reject H0

large-sample z-approximation F-test (comparative reference), and the paired bootstrap BCa test (robustness check). The F-test is included solely for comparability with traditional independent-sample methods, despite not accounting for the paired nature of the data.

Across all configurations, the null hypothesis $H_0 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} \geq 1$ was rejected with overwhelming statistical significance. For the Pitman-Morgan and F-tests, p-values fell below the numerical precision of the machine and are reported as $p < 10^{-300}$. For the paired bootstrap BCa test, the smallest possible p-value is limited by the number of resamples (B), with $p_{\min} = 1/B$; here, $B = 5000$ yields $p_{\min} = 0.0002$, which explains why the reported value is not as extreme as in the parametric tests.

Negative PM-statistics occur because the variance of the calibrated errors is consistently smaller than that of the uncalibrated errors. This sign change arises from the fact that the Pitman–Morgan statistic is proportional to the correlation between the sums and differences of paired observations, which becomes negative when the first variance is lower than the second.

The *paired bootstrap BCa* approach builds CIs without assuming normality, making it robust to heavy-tailed or skewed error distributions. Its "paired" design preserves the intrinsic correlation between $\varepsilon_{\text{calib}}$ and $\varepsilon_{\text{fast}}$ by resampling event pairs jointly. The intervals obtained matched the conclusions from the parametric tests, reinforcing the robustness of the variance reduction findings.

It is important to note that the variance ratio statistic shown for the bootstrap test (e.g., 0.3230) is identical to the F-statistic. This is not an inconsistency but rather a fundamental feature of the bootstrap methodology. The bootstrap procedure uses the original sample's calculated statistic (in this case, $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$) as its starting point. The difference between the tests lies not in this initial value, but in the method of inference. The F-test derives its p-value from a theoretical probability distribution, while the bootstrap derives it empirically by resampling the observed data. The convergence of both methods toward the same conclusion, despite their radically different assumptions, underscores the robustness and veracity of the finding.

For `BDT no-bin: PCA-10`, the variance after calibration is approximately 32% of the uncalibrated variance, corresponding to a 68% reduction. Similar reductions are observed for `BDT no-bin: CORR-90` (23%), `BDT no-bin: MUTI-80` (19%), and `BDT no-bin: MUTI-50` (23%), confirming that the improvement in resolution is consistent across all BDT configurations.

3) Statistical Interpretation: The joint occurrence of negative μ_d and variance ratios well below unity indicates that the calibration achieves both a systematic bias reduction (mean residuals closer to zero) and a variance reduction (tighter residual dispersion) relative to the uncalibrated baseline. From a statistical standpoint, these two effects act synergistically to reduce the mse of the reconstructed energy [28], [36]

$$\text{mse} = \text{Var}(\varepsilon) + [\text{Bias}(\varepsilon)]^2.$$

Since both $\text{Var}(\varepsilon)$ and $|\text{Bias}(\varepsilon)|$ are smaller after calibration, the overall MSE is necessarily reduced. This implies that the reconstructed energy distributions become both narrower and more accurately centered around the true value.

From a physics perspective, such simultaneous improvements in bias and resolution enhance the calorimeter's resolving power, benefiting tasks such as resonance peak identification, suppression of background fluctuations, and precision cross section measurements, where small biases and high resolution are critical to achieving optimal sensitivity.

F. Impact of Pile-Up on Calibration Performance

The effect of pile-up, i.e., multiple proton-proton interactions occurring within the same bunch crossing, is a well-known challenge in calorimetric energy measurements. Increasing pile-up leads to additional energy deposits unrelated to the primary hard-scatter process, which can bias the reconstructed transverse energy and degrade the resolution. In

high-luminosity scenarios, the average number of interactions per bunch crossing ($\langle\mu\rangle$) can reach several tens, exacerbating this effect.

Studies performed in [10], [23], [34], and [35] have shown that pile-up contamination in the calorimeter impacts both the bias and the variance of energy estimates, especially for low-transverse-energy objects. In the context of our proposed ML-based calibration, the use of compact and decorrelated feature sets (obtained via correlation, mutual information, and component analysis techniques) may help mitigate pile-up effects by reducing sensitivity to noisy or redundant inputs. Although this work has not explicitly quantified performance as a function of $\langle\mu\rangle$, future studies will address this by evaluating calibration accuracy under varying pile-up conditions, thereby assessing robustness for operation in the High-Luminosity LHC environment.

IV. Conclusion

This work presented a comprehensive evaluation of ML-based energy calibration methods aimed at improving the fast transverse energy estimation in HEP experiments. Using the Lorenzetti simulator and multiple feature-selection and extraction strategies, namely, correlation, mutual information ranking, PCA, and ICA, we trained and tested models to predict a calibration factor α_{calib} and correct systematic biases in the standard FastCalo measurement.

The best-performing configuration, consisting of gradient BDTs without binning and using the ten components from a PCA, achieved a reduction of up to 58.44% in RMSE and 35.22% in IQR relative to the uncalibrated baseline while maintaining compact input sets that minimize inference time.

Four top-ranked configurations, gradient BDTs without binning using ten principal components, gradient BDTs without binning using the 90 most correlated features, gradient BDTs without binning using the 50 highest mutual-information features, and gradient BDTs without binning using two independent components, all delivered prediction latencies below 10 ms on a standard laptop. This suggests that deployment on high-performance computing infrastructures, typical of modern HEP trigger systems, may allow sub-millisecond execution. In addition, the principal-component-based configuration lowered the minimum energy threshold required to sustain high event-acceptance rates, reinforcing its suitability for real-time event filtering.

Beyond point estimates, rigorous statistical validation confirmed that the observed improvements are not due to random fluctuations. Paired hypothesis tests on the mean residuals (μ_d) and variance ratios ($\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$) consistently rejected the null hypotheses with overwhelming significance, both for parametric and non-parametric methods. The joint occurrence of negative μ_d and variance ratios well below unity indicates simultaneous bias and resolution improvements, implying a substantial reduction in mse and resulting in narrower, more accurately centered reconstructed energy distributions.

These results demonstrate that ML-based calibration can enhance both the precision and operational efficiency of fast calorimetric measurements, satisfying the stringent latency constraints of trigger-level processing.

Furthermore, as the proposed methodology relies solely on high-level calorimetric observables and statistical feature extraction, it can be applied using the Lorenzetti Shower framework to different detector geometries and readout technologies without requiring extensive experiment-specific adjustments. In this sense, various detector designs are currently being integrated with the Lorenzetti framework, including next-generation detectors under development for the future circular collider (FCC) [57], [58]. The intrinsic flexibility of the used framework enables the same ML-based calibration strategy to be adapted to other HEP environments where fast and accurate energy estimation is crucial.

Future work will extend this framework to segmentation-aware models and validation with experimental data, paving the way for robust deployment in HEP data acquisition pipelines.

Appendix

The following repositories are publicly available and contain the source code and datasets related to the Lorenzetti framework used in this work.

1) Lorenzetti main repository [16]: https://github.com/lorenzetti-hep/lorenzetti
2) Dataset repository used in this study [22]: https://github.com/PauloRASpro/Lorenzetti-Shower-Simulation

These repositories include installation instructions, simulation configurations, and the datasets employed in the calibration studies described in this article.

References

[1] D. J. Griffiths, *Introduction To Elementary Particles*, 2nd ed., Hoboken, NJ, USA: Wiley, 2008.

[2] "Evidence for Higgs boson decays to a low-mass dilepton system, and a photon in,"*Phys. Lett. B*, vol. 819, Jun. 2021, Art. no. 136412.

[3] *Public ATLAS Luminosity Results for Run-3 of the LHC*. Accessed: Jul. 7, 2024. [Online]. Available: https://twiki.cern.ch/twiki/bin/view/AtlasPublic/LuminosityPublicResultsRun3

[4] ATLAS Collaboration. (2024). *Preliminary Analysis of the Luminosity Calibration for the ATLAS 13.6 TeV Data Recorded in 2023*. [Online]. Available: https://atlas.web.cern.ch/Atlas/GROUPS/PHYSICS/PUBNOTES/ATL-DAPR-PUB-2024-001

[5] H. M. Gray, "Future colliders for the high-energy frontier," *Rev. Phys.*, vol. 6, Jun. 2021, Art. no. 100053.

[6] L. Evans and P. Bryant, "LHC machine," *J. Instrum.*, vol. 3, no. 8, Aug. 2008, Art. no. S08001.

[7] O. Brüning, H. Burkhardt, and S. Myers, "The Large Hadron Collider," *Prog. Part. Nucl. Phys.*, vol. 67, no. 3, pp. 705–734, 2012.

[8] G. E. Orellana and ATLAS Collaboration. (2021). *Projected ATLAS Electron and Photon Trigger Performance in Run 3*. [Online]. Available: https://cds.cern.ch/record/2730417

[9] R. Wigmans, *Calorimetry: Energy Measurement in Particle Physics* (International series of monographs on physics). Oxford, U.K.: Clarendon, 2017.

[10] "Electron," *J. Instrum.*, vol. 14, no. 12, pp. 1–16, Dec. 2019. [Online]. Available: https://iopscience.iop.org/article/10.1088/1748-0221/14/12/P12006

[11] E. F. Simas Filho et al., "Ring-like calorimeter information for energy calibration in electron trigger at a highly segmented detector using gradient boosted decision trees," *J. Instrum.*, vol. 20, no. 6, Jun. 2025, Art. no. P06051, doi: 10.1088/1748-0221/20/06/p06051.

[12] P. Baldi, P. Sadowski, and D. Whiteson, "Searching for exotic particles in high-energy physics with deep learning," *Nature Commun.*, vol. 5, no. 1, p. 4308, Jul. 2014. [Online]. Available: https://www.nature.com/articles/ncomms5308

[13] K. Albertsson et al., "Machine learning in high energy physics community white paper," *J. Phys., Conf. Ser.*, vol. 1085, Aug. 2018, Art. no. 022008. [Online]. Available: https://iopscience.iop.org/article/10.1088/1742-6596/1085/2/022008

[14] J. Hoya, "The ATLAS electron and photon trigger in Run-2," in *Proc. PoS(LHCP)*, 2018, p. 27. [Online]. Available: https://cds.cern.ch/record/2637126/files/ATL-DAQ-PROC-2018-020.pdf

[15] M. V. Araujo et al., "Lorenzetti showers–A general-purpose framework for supporting signal reconstruction and triggering with calorimeters," *Comput. Phys. Commun.*, vol. 286, May 2023, Art. no. 108671. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0010465523000164

[16] J. V. da Fonseca Pinto, jlieberm, W. Freund, G. G. Milan, M. V. de Araujo, *Lorenzetti-Hep/Lorenzetti: Version 2.0.0*. [Online]. Available: https://github.com/lorenzetti-hep/lorenzetti

[17] G. Aad et al., "The ATLAS trigger system for LHC run 3 and trigger performance in 2022," *J. Instrum.*, vol. 19, no. 6, Jun. 2024, Art. no. P06029, doi: 10.1088/1748-0221/19/06/p06029.

[18] Geant Collaboration. *Geant4 User's Guide for Appl. Developers*. (Sep. 2017). [Online]. Available: https://indico.cern.ch/event/647154/contributions/2714212/attachments/1529029/2397032/BookForApplicationDevelopers.pdf

[19] C. Bierlich et al., "A comprehensive guide to the physics and usage of PYTHIA 8.3," *SciPost Phys. Codebases*, vol. 8, p. 8, Nov. 2022, doi: 10.21468/SciPostPhysCodeb.8.

[20] T. M. Quirino, T. C. A. Paschoalin, G. I. Gonçalves, P. H. B. Lisboa, L. M. de Andrade Filho, and B. S.-M. Peralva, "Online pulse compensation for energy spectrum determination: A pole-zero cancellation and unfolding approach," *Electronics*, vol. 14, no. 3, p. 493, Jan. 2025.

[21] L. Boggia and B. Malaescu, "Synthetic data generation with lorenzetti for time series anomaly detection in high-energy physics calorimeters," 2025, *arXiv:2509.07451*.

[22] P. R. A. da Silva and E. F. de Simas Filho. (2025). *Lorenzetti Shower Simulation Dataset*. [Online]. Available: https://github.com/PauloRASpro/Lorenzetti-Shower-Simulation

[23] G. Aad et al., "Electron and photon energy calibration with the ATLAS detector using LHC run 2 data," *Eur. Phys. J. C*, vol. 80, p. 1105, 2023. [Online]. Available: https://iopscience.iop.org/article/10.1088/1748-0221/19/02/P02009

[24] M Aaboud et al., "Electron and photon energy calibration with the ATLAS detector using 2015–2016 LHC proton-proton collision data," *J. Instrum.*, vol. 14, no. 3, pp. 1–17, Mar. 2019. [Online]. Available: https://iopscience.iop.org/article/10.1088/1748-0221/14/03/P03017

[25] J. V. da Fonseca Pinto, "Ring-shaped calorimetry information for a neural egamma identification with ATLAS detector," Geneva, Switzerland, Tech. Rep. ATL-DAQ-PROC-2016-007, 2016. [Online]. Available: https://cds.cern.ch/record/2142761

[26] J. M. Seixas, L. P. Caloba, M. N. Souza, A. L. Braga, and A. P. Rodrigues, "Neural second-level trigger system based on calorimetry," *Comput. Phys. Commun.*, vol. 95, nos. 2–3, pp. 143–157, Jun. 1996. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0010465596000124

[27] S. S. Haykin, *Neural Networks and Learning Machines*, 3rd ed., Upper Saddle River, NJ, USA: Pearson, 2009.

[28] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Cham, Switzerland: Springer, 2009.

[29] A. Hyvärinen, J. Karhunen, and E. Oja, *Independent Component Analysis*. Hoboken, NJ, USA: Wiley, 2001. [Online]. Available: https://onlinelibrary.wiley.com/doi/book/10.1002/0471221317

[30] A. Papoulis and S. U. Pillai, *Probability, Ramdom Variables and Stochastic Process*, 4th ed., New York, NY, USA: McGraw-Hill, 2002.

[31] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. Hoboken, NJ, USA: Wiley, 1991.

[32] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3146–3154.

[33] L. de Oliveira, M. Kagan, L. Mackey, B. Nachman, and A. Schwartzman, "Jet-images—Deep learning edition," *J. High Energy Phys.*, vol. 2016, no. 7, pp. 1–31, Jul. 2016. [Online]. Available: https://link.springer.com/article/10.1007/JHEP07(2016)069

[34] CMS Collaboration, "Performance of electron reconstruction and selection with the CMSdetector in proton-proton collisions at $\sqrt{s}$ = 8 TeV," *J. Instrum.*, vol. 10, no. 6, p. 06005, 2015. [Online]. Available: http://dx.doi.org/10.1088/1748-0221/10/06/P06005

[35] T. Dimova, "Calibration and performance of the CMS electromagnetic calorimeter during the LHC run II," CERN, Geneva, Switzerland, Tech. Rep. CMS-CR-2018-377, 2018. [Online]. Available: https://cds.cern.ch/record/2668251

[36] G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed., Pacific Grove, CA, USA: Duxbury, 2002.

[37] D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, 6th ed., Hoboken, NJ, USA: Wiley, 2014.

[38] R. V. Hogg, J. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, 8th ed., Boston, MA, USA: Pearson, 2019.

[39] E. L. Lehmann and J. P. Romano, *Test. Stat. Hypotheses*, 3rd ed., New York, NY, USA: Springer, 2005.

[40] A. M. Mood, F. A. Graybill, and D. C. Boes, *Introduction to the Theory of Statistics*, 3rd ed. New York: McGraw-Hill, 1974.

[41] R. B. D'agostino, "An omnibus test of normality for moderate and large size samples," *Biometrika*, vol. 58, no. 2, pp. 341–348, 1971. [Online]. Available: https://www.jstor.org/stable/2334522

[42] R. B. D'Agostino and E. S. Pearson, "Tests for departure from normality. empirical results for the distributions of b_2 and $\sqrt{b_1}$," *Biometrika*, vol. 60, no. 3, pp. 613–622, 1973. [Online]. Available: https://www.jstor.org/stable/2335012

[43] T. W. Anderson and D. A. Darling, "A test of goodness of fit," *J. Amer. Stat. Assoc.*, vol. 49, no. 268, pp. 765–769, 1954. [Online]. Available: https://www.jstor.org/stable/2281537

[44] N. M. Razali and Y. B. Wah, "Power comparisons of Shapiro–Wilk, Kolmogorov–Smirnov, Lilliefors and anderson–darling tests," *J. Stat. Model. Analytics*, vol. 2, no. 1, pp. 21–33, 2011. [Online]. Available: https://www.researchgate.net/publication/267205556

[45] M. A. Stephens, "EDF statistics for goodness of fit and some comparisons," *J. Amer. Stat. Assoc.*, vol. 69, no. 347, pp. 730–737, Sep. 1974. [Online]. Available: https://www.jstor.org/stable/2286009

[46] H. C. Thode, *Testing for Normality*. New York, NY, USA: Marcel Dekker, 2002.

[47] D. S. Moore and G. P. McCabe, *Introduction to the Practice of Statistics*, 5th ed., New York, NY, USA: W. H. Freeman, 2006.

[48] T. Lumley, P. Diehr, S. Emerson, and L. Chen, "The importance of the normality assumption in large public health data sets," *Annu. Rev. Public Health*, vol. 23, no. 1, pp. 151–169, May 2002. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/11910059/

[49] J. A. Rice, *Mathematical Statistics and Data Analysis*. Belmont, CA, USA: Duxbury Press, 2006.

[50] M. P. Fay and M. A. Proschan, "Wilcoxon-mann-whitney or t-test? On assumptions for hypothesis tests and multiple interpretations of decision rules," *Statist. Surveys*, vol. 4, no. none, pp. 1–39, Jan. 2010. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/20414472/

[51] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, p. 80, 1945. [Online]. Available: https://www.jstor.org/stable/3001968

[52] E. J. G. Pitman, "A note on normal correlation," *Biometrika*, vol. 31, no. 1/2, pp. 9–12, Jul. 1939. [Online]. Available: https://www.jstor.org/stable/2334971

[53] W. A. Morgan, "A test for the significance of the difference between the two variances in a sample from a normal bivariate population," *Biometrika*, vol. 31, no. 1/2, pp. 13–19, Jul. 1939. [Online]. Available: https://www.jstor.org/stable/2334972

[54] B. Efron, "Bootstrap methods: Another look at the jackknife," *Ann. Statist.*, vol. 7, no. 1, pp. 1–26, Jan. 1979. [Online]. Available: https://www.jstor.org/stable/2958830

[55] A. C. Davison and D. V. Hinkley, *Bootstrap Methods and Their Application*. Cambridge, U.K.: Cambridge Univ. Press, 1997.

[56] R. Wigmans, *Calorimetry: Energy Measurement in Particle Physics*. London, U.K.: Oxford Univ. Press, 2017.

[57] I. Agapov et al., "Future circular lepton collider FCC-ee: Overview and status," 2022, *arXiv:2203.08310*.

[58] *The IDEA Study Group*, "The idea detector concept for fcc-ee," 2025. [Online]. Available: https://doi.org/10.48550/arXiv.2502.21223

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - ROR identifier: 00x0ma614